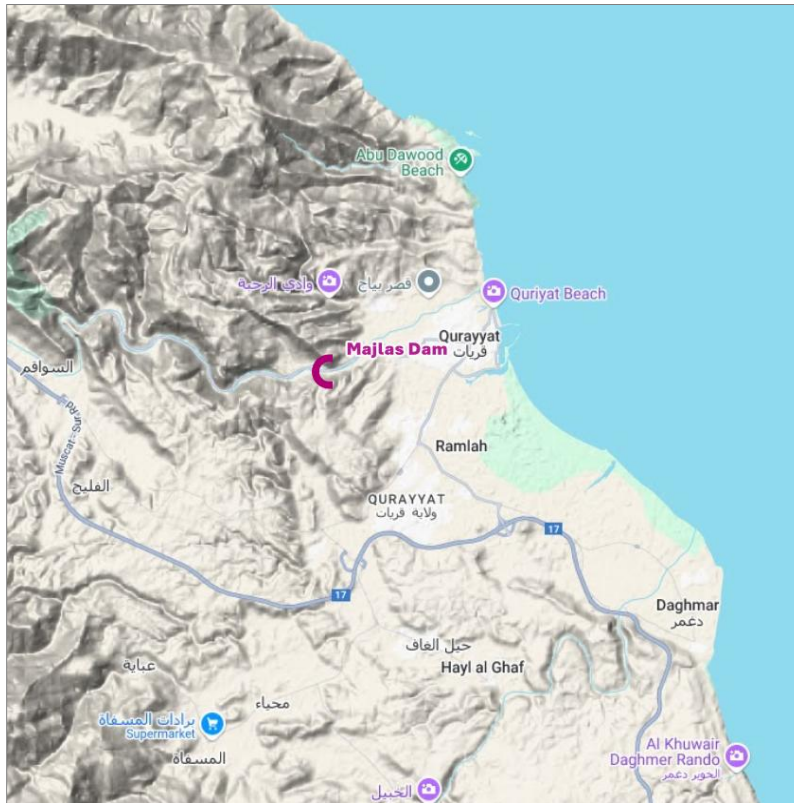




Sultanate of Oman

Ministry of Agriculture, Fisheries Wealth & Water Resources

Consultancy Services for Design Review and Supervision of Wadi Majlas Dam in Wilayat Qurayat, Muscat Governorate



Wadi Majlas Flood Protection Dam Dam Break Analysis Methodology Report

Rev 00

September 2026



RENARDET S.A. & Partners
CONSULTING ENGINEERS L.L.C.



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ABBREVIATIONS AND NOMENCLATURE

The following abbreviations, symbols and units are used throughout this report.

Table 1 - Abbreviations and nomenclature.

Term	Meaning
2D	Two-dimensional. A model that computes flow depth and velocity over a grid of cells covering the ground, not only along a line.
AIDR	Australian Institute for Disaster Resilience, publisher of the flood hazard classes H1 to H6 used in this study.
ANCOLD	Australian National Committee on Large Dams, whose 2012 guideline gives the consequence categories for dams.
D×V	Depth multiplied by velocity (m ² /s), the usual measure of how dangerous flowing water is to people and buildings.
DTM / DSM	Digital terrain model (bare ground) / digital surface model (ground plus buildings and trees).
EAP	Emergency action plan: the document that says who does what when the dam is in danger.
EAV	Elevation-area-volume relationship of the reservoir.
EWS	Early warning system.
FEMA	Federal Emergency Management Agency (United States), publisher of the federal dam-safety guidelines.
FERC	Federal Energy Regulatory Commission (United States), whose engineering guidelines give breach parameters by dam type.
FSL	Full supply level, 84.5 m a.s.l.; the spillway crest level.
GERCC	Grout-enriched roller-compacted concrete, the watertight facing of the dam.
GHS-POP	Global Human Settlement population grid (European Commission), 100 m cells, 2025 edition.
HEC-RAS	River Analysis System of the US Army Corps of Engineers Hydrologic Engineering Center; the hydraulic model used.
ICOLD	International Commission on Large Dams.
IDF	Inflow design flood: the flood the dam must pass safely; here the PMF.
LOL	Loss of life (estimated number of fatalities).
MAFWR	Ministry of Agriculture, Fisheries Wealth and Water Resources, the Client.
MWL	Maximum water level in the reservoir during the inflow design flood (96.0 m in the design table, 96.7 m on the dam section).
NSA	National Survey Authority of Oman, source of the 2 m and 5 m national terrain models.
PAR	Population at risk: the people inside the flooded area.
PMF / PMP	Probable maximum flood / probable maximum precipitation.
RCC	Roller-compacted concrete.
RCEM	Reclamation Consequence Estimating Methodology (US Bureau of Reclamation, 2015) for loss-of-life estimates.
SWE	Shallow water equations, the full set of flow equations solved by the 2D model.
USACE / USBR	US Army Corps of Engineers / US Bureau of Reclamation.
m a.s.l.	Metres above sea level (project datum).
Mm ³	Million cubic metres.



Term	Meaning
m ³ /s	Cubic metres per second.



EXECUTIVE SUMMARY

This report sets out how the dam break analysis of the Wadi Majlas Flood Protection Dam will be carried out. It is a methodology, not a result: it fixes the failure scenarios, the breach parameters, the hydraulic model, the maps, the consequence estimates and the quality checks before any model is run, so that the Client can approve the basis first and the later results cannot be questioned on method.

The dam, as finalised in the design review of September 2026, is a roller-compacted concrete (RCC) gravity dam, 72.5 m above the wadi bed and 312 m long at the crest, with a free ogee spillway 200 m wide on the dam body and a 102.2 Mm³ reservoir at full supply level. Qurayat town starts about 2 km downstream.

The single most important decision in this report follows from the dam type. A concrete gravity dam does not fail the way an earth or rockfill dam fails. An embankment is eroded away by the water over hours, and its breach size and timing are estimated with regression equations fitted to past embankment failures. A gravity dam fails when one or more of its blocks, called monoliths, slide or overturn on the foundation. That happens within minutes, the opening has vertical sides, and its width is a whole number of blocks. The regression equations used in earlier studies of embankment dams in Oman therefore do not apply here, and this methodology replaces them with the monolith-based approach of the United States federal guidance for concrete dams, with the number of failed blocks varied to cover the uncertainty.

Three scenarios are analysed, each with and without the downstream training dikes: a sunny-day failure with the reservoir at full supply level (84.5 m) and no flood; a flood-day failure at the peak pool of the probable maximum flood (96.7 m); and the probable maximum flood passing the dam without failure, which is the baseline needed to separate the damage caused by the dam failing from the damage the flood would have caused anyway. Sensitivity runs vary the breach width, the breach time, the ground roughness, the mesh size, the sea level and the starting pool.

The flood is simulated in HEC-RAS 2D with the full shallow water equations, on a model that covers the reservoir, the dam and the wadi down to the Gulf of Oman. The dam is represented as a connection between the reservoir and the downstream area, carrying the crest, the spillway and the



breach. The existing HEC-RAS model of the downstream reach, built for the flood protection design, is reused and extended.

The outputs are maps of maximum depth, velocity, depth times velocity, arrival time and duration, the AIDR hazard classes H1 to H6 already used in the flood-risk study of the scheme, and for each scenario the population at risk, an estimate of loss of life by the USBR method, and the economic damage. From these the dam receives a consequence category and a hazard class, and the emergency action plan receives its warning times and evacuation maps.

Four items are needed from the design team before the model is built: the monolith joint layout with block widths, the routed maximum water level during the PMF (96.0 or 96.7 m), the elevation-area-volume table for the straight axis, and the design surfaces of the dam, stilling basin and dikes. Table 2 summarises the analysis at a glance.

Table 2 - The dam break analysis at a glance.

Item	Decision in this methodology
Dam type and failure model	RCC gravity dam; failure by sliding of whole monoliths at the foundation; breach with vertical sides formed in about 0.1 h
Breach width	Base case two spillway bays (86 m); lower case one bay (43 m); upper case half the crest length (156 m)
Scenarios	Sunny day at FSL 84.5 m; flood day at the PMF pool 96.7 m; PMF without failure (baseline); each with and without the dikes
Hydraulic model	HEC-RAS 2D, version 6.6, full shallow water equations, reservoir and downstream 2D areas joined by the dam connection, sea boundary
Terrain	5 cm dam-site survey, 2 m drone DTM, 5 m NSA DTM and the design surfaces of dam, basin and dikes, merged in that order of priority
Map products	Maximum depth, velocity, $D \times V$, arrival time, duration, AIDR hazard class H1 to H6, incremental depth; 2 m rasters; map series in the project layout
Consequences	Population at risk from GHS-POP 2025; loss of life by USBR RCEM (2015); damage with the flood-risk study curves; incremental over the no-failure case
Classification	ANCOLD (2012) consequence category and FEMA hazard potential class
Sensitivity	Breach width, breach time, roughness, mesh size, sea level, starting pool
Quality checks	Volume balance, hand check of the peak outflow, stability, comparison of the no-failure routing with the design review



1. INTRODUCTION

1.1. PURPOSE OF THIS REPORT

The Ministry of Agriculture, Fisheries Wealth and Water Resources is building a flood protection dam on Wadi Majlas, upstream of Qurayat in Muscat Governorate. Renardet S.A. & Partners is the consultant for the design review and supervision. The scope of services includes a dam break analysis¹ whose results feed the dam classification, the emergency action plan and the early warning system.

This report is the methodology for that analysis. It is issued before the analysis so that the Client can approve the basis: the scenarios, the breach parameters, the model, the maps, the consequence method and the checks. Once approved, it becomes the reference for the analysis report that follows, and any later change to the basis will be recorded against it.

1.2. WHY A METHODOLOGY COMES FIRST

Two features of this project make an agreed methodology more important than usual. First, the dam is a concrete gravity dam, while most dam break studies in the region, and the standard software guidance, are written around embankment dams. The failure mechanism and the breach parameters are different, and a study that applied embankment rules to a gravity dam would give the wrong size of flood and the wrong warning times. Second, the dam stands only about 2 km upstream of Qurayat, so the consequences of the choice of parameters are large, and the parameters must be set and defended in the open rather than left to the modeller.

1.3. WHAT CHANGES FROM EARLIER PRACTICE

Earlier dam break studies for the Client, such as the Al Dreez dam study of 2025, followed the HEC-RAS guidance for estimating breach parameters and analysed a sunny-day and a rainy-day scenario

¹ A dam break analysis is a study that assumes the dam fails, computes the flood wave that leaves the reservoir, follows it downstream with a hydraulic model, and reports where the water goes, how deep and fast it is, when it arrives, and who and what is in its path. It does not say whether the dam will fail; it says what would happen if it did.



for a rockfill dam. That approach is sound for an embankment dam. Table 3 lists what is kept from that practice and what is changed here, and why.

Table 3 - What is kept and what changes from the earlier dam break practice.

Item	Earlier practice (embankment dams)	This methodology (RCC gravity dam)	Reason
Breach shape and size	Trapezoid from regression equations (Froehlich, MacDonald, Von Thun, Xu and Zhang)	Rectangular opening equal to whole monoliths, vertical sides; width set from the block layout	Regression equations were fitted to earth and rockfill failures and do not apply to concrete dams
Breach time	0.5 to several hours from the equations	About 0.1 h (6 minutes), range 0.05 to 0.3 h	A sliding monolith leaves the dam in minutes; the federal guidance gives 0.1 to 0.5 h for gravity dams
Sunny-day cause	Piping through the embankment	Sliding of blocks on the foundation, for example after an earthquake or a loss of uplift control	There is no soil to pipe through an RCC body
Flood-day cause	Overtopping and erosion of the embankment	Failure of blocks at the highest pool of the PMF, while the spillway is running	RCC gravity dams tolerate overtopping; the loading on the foundation is what matters
Scenarios	Sunny day and rainy day	Sunny day, flood day and the PMF without failure; each with and without the dikes	The no-failure run is needed to compute the incremental effect of the failure, which decides the classification
Model	HEC-RAS 2D	HEC-RAS 2D, full shallow water equations, reservoir modelled in 2D	Kept; the reservoir in 2D reproduces the sudden drawdown of an instant breach
Consequences	Inundation and hazard maps	Maps plus population at risk, loss of life, damage, consequence category	Required for the classification, the emergency plan and the warning system
Sensitivity	Largest breach width from the equations	Named runs on width, time, roughness, mesh, sea level, pool	The uncertainty must be shown, not hidden in one conservative value

1.4. HOW TO READ THIS REPORT

Chapter 2 describes the dam, the reservoir, the reach downstream and the data in hand. Chapter 3 lists the standards followed and how they rank. Chapter 4 explains the kinds of dams, how each kind breaks, and which rules apply to Majlas. Chapter 5 defines the failure scenarios and Chapter 6 the breach parameters. Chapter 7 is the hydraulic model, written as a step-by-step guide for HEC-RAS, and Chapter 8 the maps and hazard classes. Chapter 9 covers the consequences and the dam classification, Chapter 10 the sensitivity runs and quality checks, Chapter 11 the deliverables, the



work steps and the data still needed, and Chapter 12 the conclusions. Technical terms are explained in footnotes where they first appear, so that the main text stays readable.

1.5. SCOPE AND LIMITS

- The analysis assumes the dam fails. It does not assess the probability of failure and it is not a structural or geotechnical check of the design.
- The upstream Haifaz dam is not included as a cascade failure. Its catchment is 40 km² against 571 km² for Majlas, and the Majlas reservoir stores the whole 200-year flood of 90.5 Mm³ below the spillway crest, so the volume released by a Haifaz failure would be absorbed without raising the Majlas pool to a dangerous level.
- The bottom outlet is assumed closed in all scenarios. Its capacity of 120 m³/s is less than one percent of the breach flows and does not change the results.
- Results will be presented as depths, velocities, times and counts, with their sensitivity. They are estimates for planning and classification, not predictions of a particular event.



2. THE DAM, THE RESERVOIR AND THE REACH DOWNSTREAM

2.1. THE SCHEME

Wadi Majlas drains 571.3 km² of the eastern Hajar mountains and reaches the Gulf of Oman at Qurayat. The flood protection scheme has three parts: the Majlas dam in the gorge upstream of the town, the smaller Haifaz dam on a side catchment further upstream, and about 2,600 m of training dikes and bank protection through the town, sized for the 200-year flood. Figure 1 shows the catchments, the reservoir and the reach to the sea.



Wadi Majlas dam: catchments, reservoir and the reach to Qurayat

Wadi Majlas Flood Protection Dam - Dam Break Analysis Methodology | Renardet S.A. & Partners, 2026

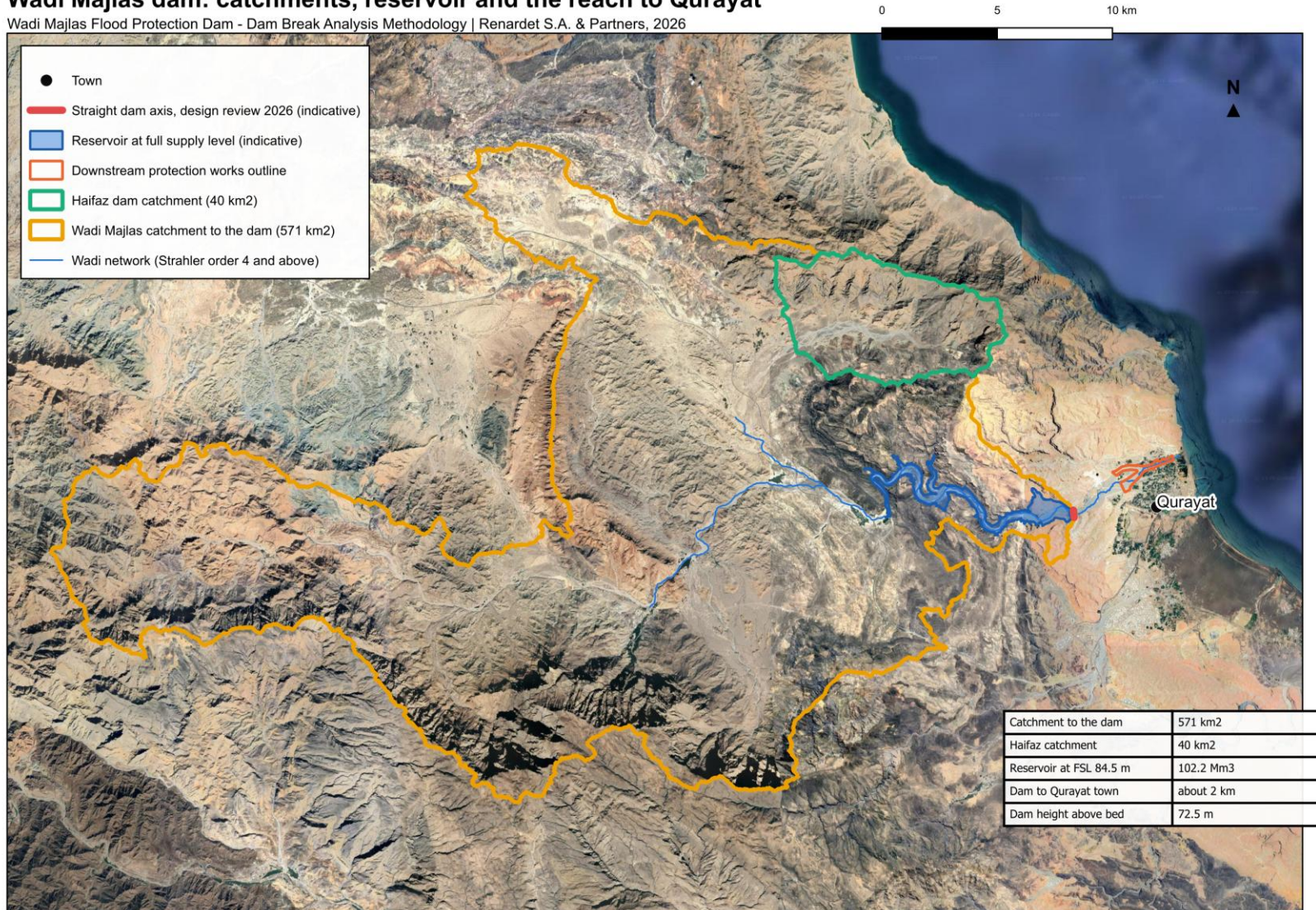


Figure 1 - Wadi Majlas dam: catchments, reservoir and the reach down to Qurayat and the sea.

2.2. THE DAM AS FINALISED

The design review changed the dam from the original curved axis to a straight axis 110 m further downstream, where the bedrock is shallower and of better quality (Figure 4). The dam is a roller-compacted concrete gravity dam with a grout-enriched RCC facing¹, 72.5 m high above the wadi bed and 110.5 m above the foundation level of -15.0 m a.s.l. The crest is at 95.5 m with a parapet to 96.7 m, 6 m wide and 312 m long. The upstream face is nearly vertical (0.05H:1V) and the downstream face slopes at 0.75H:1V. Table 4 lists the characteristics used in this study, all taken from the design review presentation of 21 September 2026, and Figure 2 and Figure 3 show the section and the general plan.

Table 4 - Characteristics of the Majlas dam used in this study.

Characteristic	Value	Source
Dam type	Roller-compacted concrete (RCC) gravity dam with grout-enriched RCC facing	Design review
Crest level / parapet top	95.5 / 96.7 m a.s.l.	Design review
Crest width / crest length	6 m / 312 m	Design review
Height above wadi bed / above foundation	72.5 m / 110.5 m	Design review
Foundation level	-15.0 m a.s.l.	Design review
Wadi bed at the axis	23.0 m a.s.l. (crest minus height above bed)	Derived
Upstream / downstream face	0.05H:1V / 0.75H:1V	Design review
Blocks (monoliths)	17 blocks B1 to B17 on the general plan; widths to be supplied	Design review drawing
Axis	Straight, 110 m downstream of the original arch axis	Design review
Foundation treatment	Grout curtain 60 m deep, drainage curtain, consolidation grouting, galleries at three levels	Design review

¹ Roller-compacted concrete is a dry concrete placed in thin layers and compacted with rollers, like a road base. It is used for gravity dams because it is fast and economical to place. The grout-enriched facing is the same concrete made richer in cement mortar at the upstream face so that it is watertight.



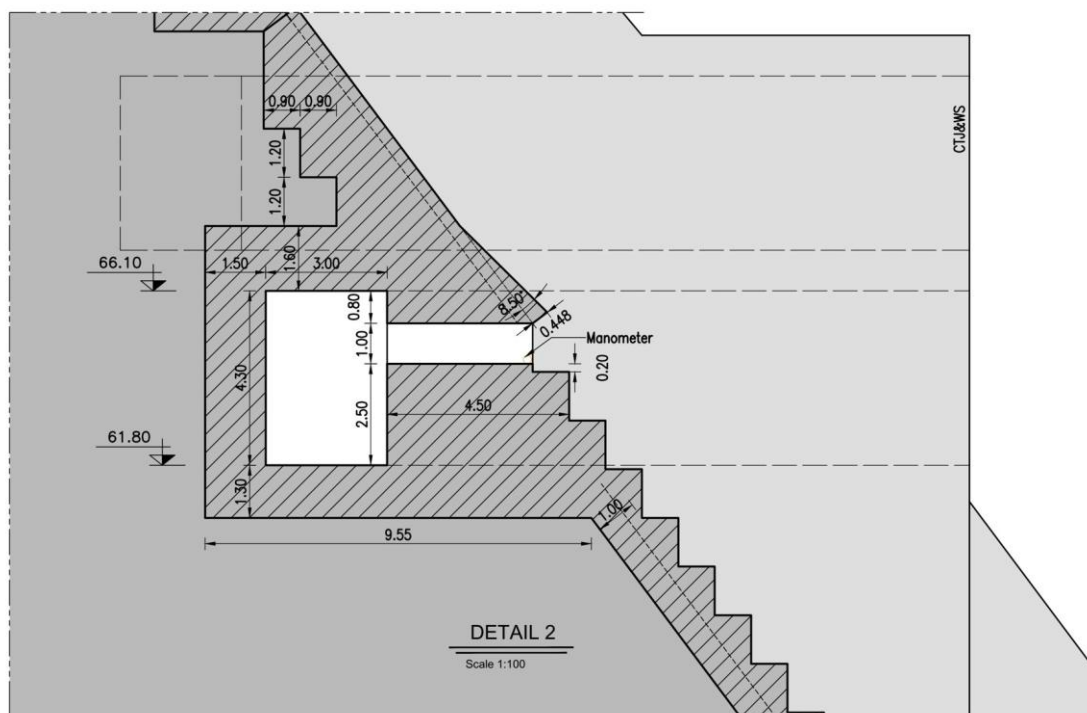


Figure 2 - Section through the spillway blocks with the full supply level (84.5 m) and the PMF level (96.7 m) marked (design review, September 2026).

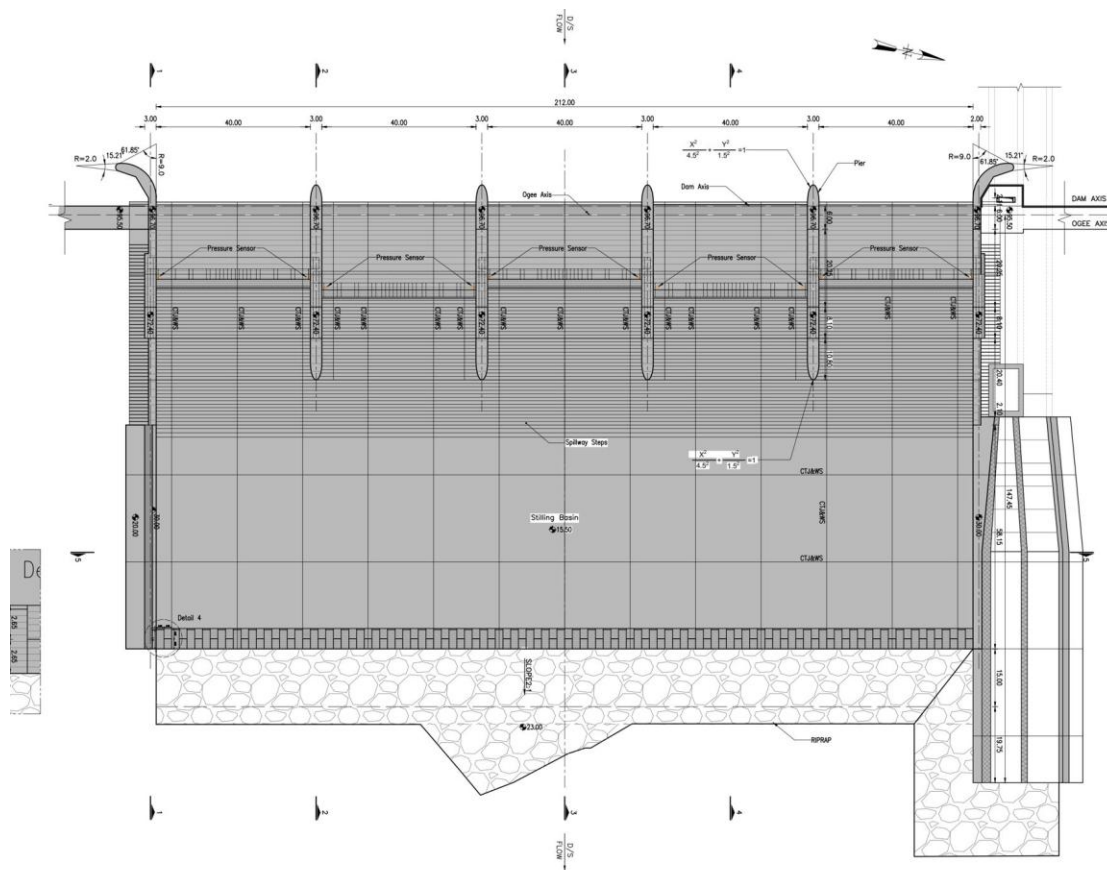


Figure 3 - General plan of the dam, spillway, stilling basin and bottom outlet on the straight axis (design review, September 2026).

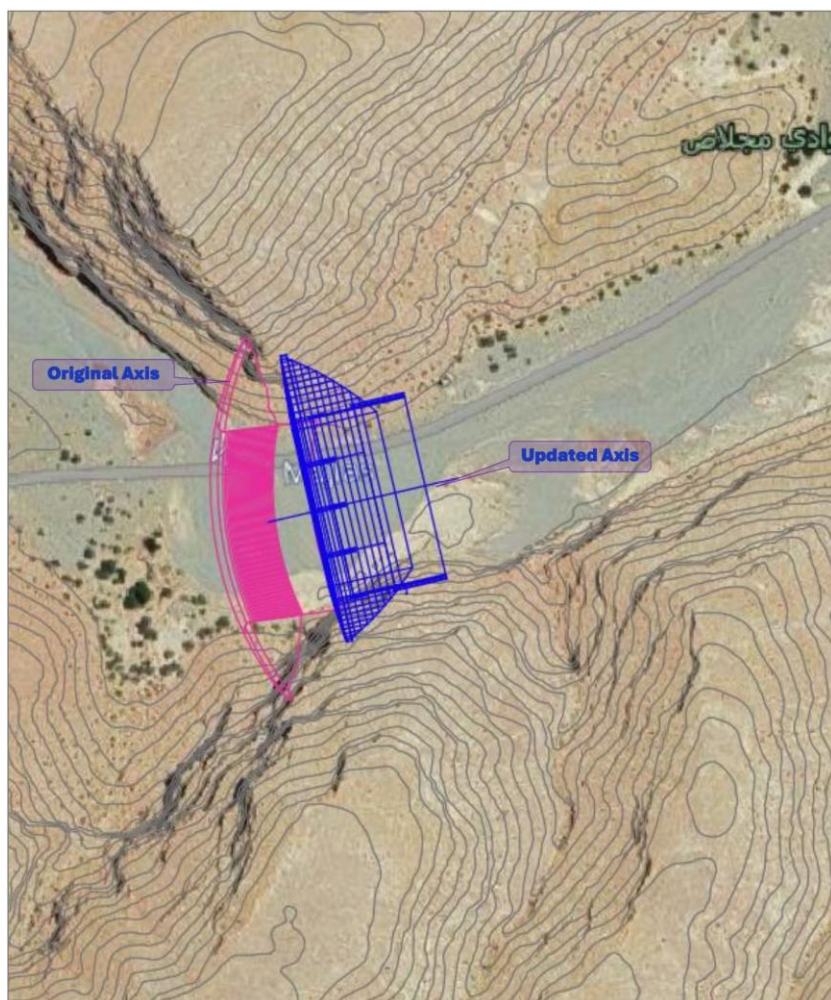


Figure 4 - Original arch axis and the updated straight axis, 110 m downstream (design review, September 2026).

2.3. SPILLWAY AND BOTTOM OUTLET

The spillway is a free overflow ogee¹ at 84.5 m, on the dam body, with 5 bays of 40 m separated by 4 piers of 3 m, a net length of 200 m and a gross length of 212 m. The flow runs down a stepped chute with aerators into a USBR Type II stilling basin 55 m long with its floor at 15.5 m (Figure 5). The design

¹ An ogee is a spillway crest shaped like the underside of a falling sheet of water, so that the flow leaves the crest smoothly. Free overflow means there are no gates: water passes as soon as the reservoir rises above the crest.

capacity is 17,000 m³/s. Figure 7 shows the rating curve¹ that reproduces this capacity, which the hydraulic model must match in the no-failure run.

The bottom outlet uses the left diversion culvert. A single steel pipe of 2.8 m diameter takes water from intakes at 36 and 44 m through a gate chamber at the downstream toe, and discharges up to 120 m³/s onto a trajectory chute into the stilling basin (Figure 6). Table 5 lists the values used.

Table 5 - Spillway and bottom outlet characteristics used in this study (design review, September 2026).

Characteristic	Value
Spillway type	Free overflow ogee with stepped chute, ungated
Crest level	84.5 m a.s.l. (= FSL)
Bays / piers	5 × 40 m / 4 × 3 m
Net / gross length	200 m / 212 m
Design capacity	17,000 m ³ /s
Unit discharge, design flood / PMF	60.7 / 85.0 m ² /s
Stilling basin	USB Type II, 55 m long, floor at 15.5 m a.s.l.
Bottom outlet	One steel pipe 2.8 m, intakes at 36 and 44 m, 120 m ³ /s
Bottom outlet gates	1.8 x 2.5 m service radial gate, 2.0 x 2.6 m emergency slide gate, 2.8 x 2.8 m bulkhead gate
Diversion (construction)	Two 6.0 × 6.5 m culverts, cofferdam crest 37.0 m; plugged after construction

¹ The rating curve of a spillway is the relation between the reservoir level and the discharge over the crest. For a free ogee it follows $Q = C L H^{1.5}$, where L is the net length and H the head above the crest.



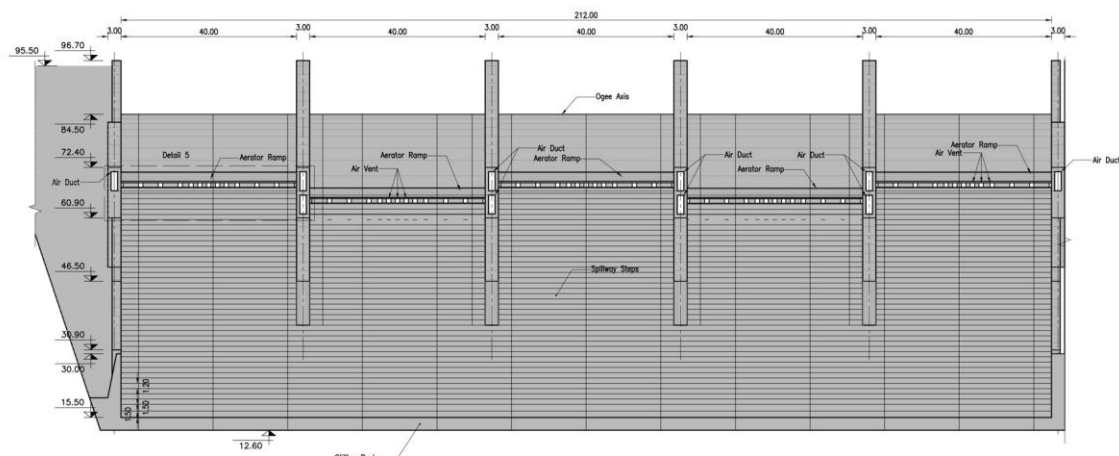


Figure 5 - Downstream view of the spillway: five bays, four piers, stepped chute and stilling basin (design review, September 2026).

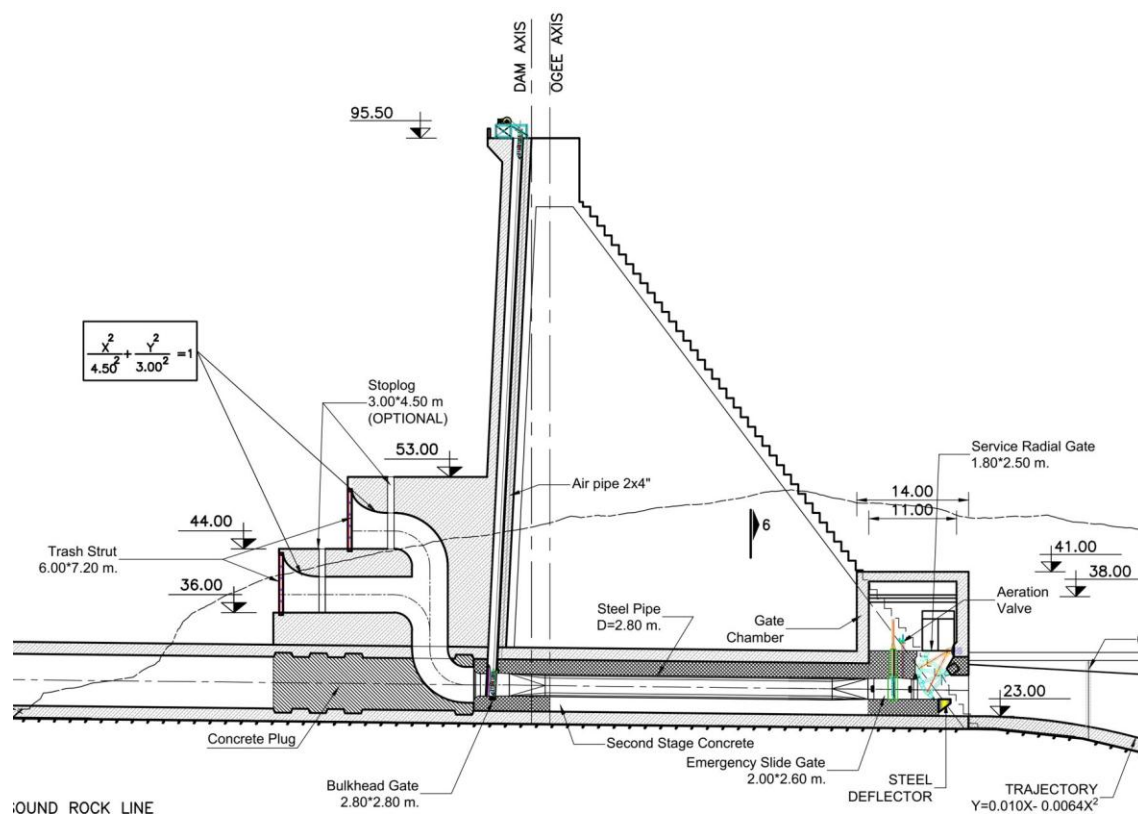


Figure 6 - Section through the bottom outlet in the left diversion culvert (design review, September 2026).

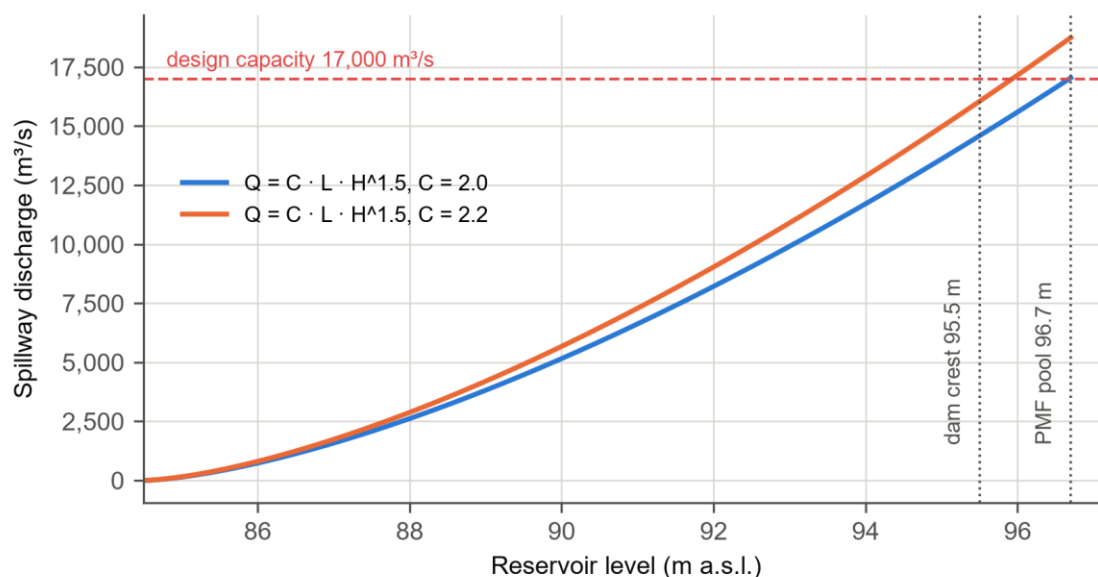


Figure 7 - Spillway rating curve for the free ogee, $Q = C L H^{1.5}$ with $L = 200$ m and C between 2.0 and 2.2 $m^{0.5}/s$; the design capacity of $17,000$ m^3/s is reached near the PMF pool.

2.4. RESERVOIR

At the full supply level of 84.5 m the reservoir holds 102.2 Mm^3 and at the maximum water level about 150.7 Mm^3 (Figure 8). The elevation-area-volume curves in hand (Figure 9) were derived for the original axis; the straight axis lies 110 m downstream and adds a small volume, so the design team is asked to supply the curve for the final axis. The curve matters twice in this study: it sets how much water is behind the dam when it fails, and it sets how fast the pool drops while the breach flows. Table 6 gives the levels used.

Two values are quoted in the design review for the highest pool during the PMF: 96.0 m in the characteristics table and 96.7 m on the dam section, which is also the parapet top. This methodology uses 96.7 m for the flood-day scenario until the design team confirms the routed value. Either way the PMF pool stands about 1.2 m above the non-overflow crest at 95.5 m and is held only by the parapet wall. A parapet is a wave wall, not a retaining element, and international practice does not credit it in dam break work; this methodology therefore treats the crest as overtopped during the PMF in the no-failure run, and runs a sensitivity case in which the parapet holds.



Table 6 - Reservoir levels and volumes used in this study.

Level	Elevation (m a.s.l.)	Storage (Mm ³)	Use in this study
Wadi bed at the dam	23.0	0	Breach bottom
Bottom outlet intakes	36 and 44	-	Closed in all scenarios
Full supply level (spillway crest)	84.5	102.2	Sunny-day pool
Dam crest	95.5	-	About 1.2 m below the PMF pool; parapet not credited
PMF pool (parapet top)	96.7	about 150.7	Flood-day pool

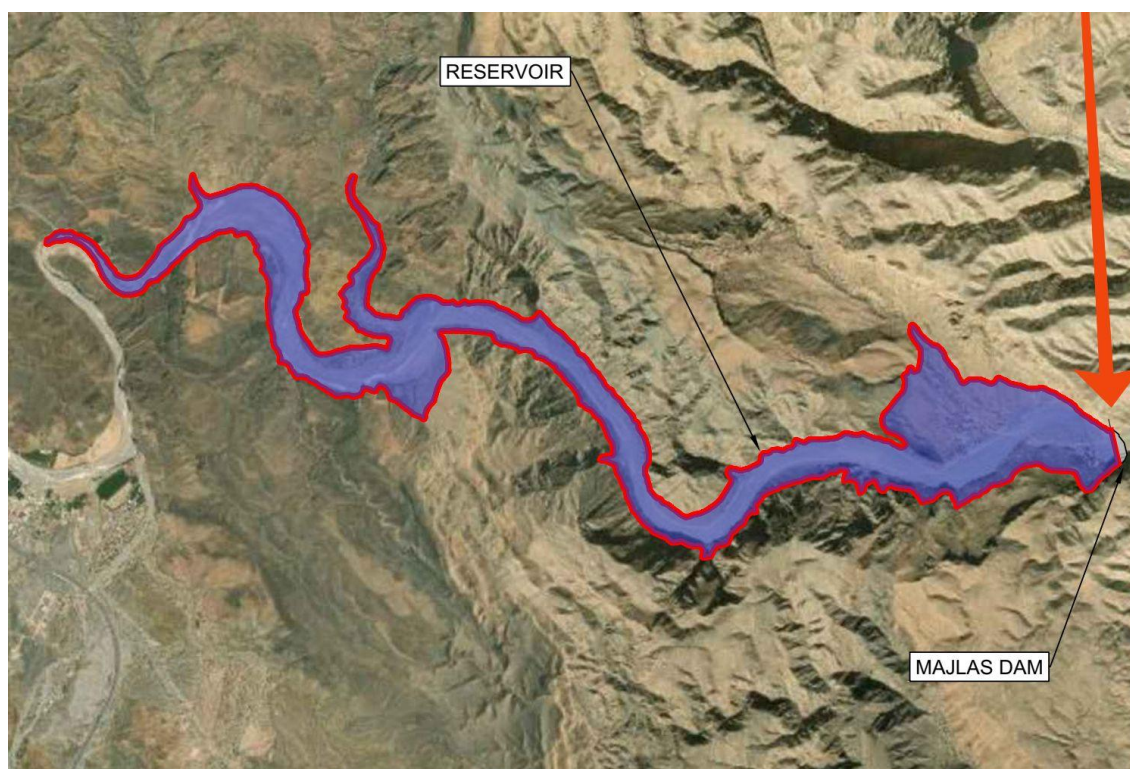


Figure 8 - Reservoir at full supply level (design review, September 2026).

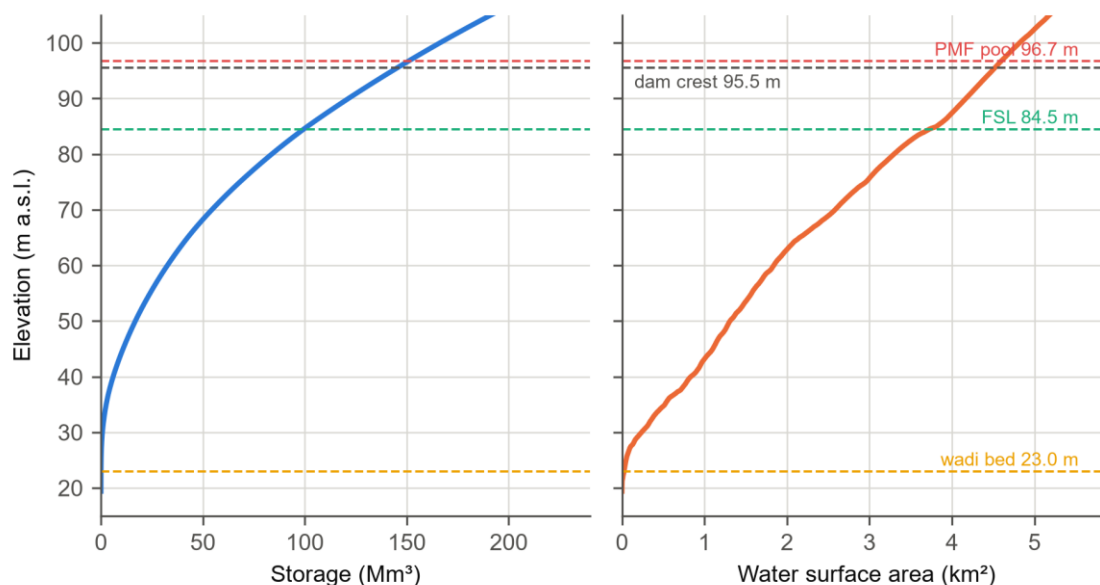


Figure 9 - Reservoir elevation-storage and elevation-area curves (original axis, hydrology report) with the levels used in this study.

2.5. DESIGN FLOODS

The hydrology report and the design review give the inflow floods at the dam (Table 7, Figure 10). The reservoir stores the whole 200-year flood of 90.5 Mm³ below the spillway crest. The PMF, with a peak of 18,179 m³/s and a volume of 549.1 Mm³, is the inflow design flood¹ and drives the flood-day scenario. Figure 11 shows the hydrograph shapes: the PMF peaks about 15 hours into the 48-hour storm, which fixes when the flood-day failure is triggered.

Table 7 - Design floods at the Majlas dam and the two largest recorded cyclones (design review, September 2026).

Return period	24-h rainfall (mm)	Peak inflow (m ³ /s)	Volume (Mm ³)
5-yr	55.3	421	11.1
10-yr	105.5	1,124	25.0
25-yr	137.5	1,805	42.8
50-yr	168.1	2,477	55.5
100-yr	205.4	3,321	71.1

¹ The inflow design flood is the largest flood the dam is designed to pass safely. For a dam whose failure could cause loss of life, international practice takes it as the probable maximum flood, the flood from the largest rainfall that is physically possible over the catchment.



Return period	24-h rainfall (mm)	Peak inflow (m ³ /s)	Volume (Mm ³)
200-yr	251.0	4,370	90.5
500-yr	327.2	6,143	123.1
1,000-yr	399.9	7,856	154.5
2,000-yr	488.7	9,115	192.9
5,000-yr	639.1	12,378	258.2
10,000-yr	685.1	13,375	278.1
PMF	1306.6	18,179	549.1
Cyclone Gonu, June 2007	501.0	6,538	265.7
Cyclone Phet, June 2010	264.0	2,036	127.3

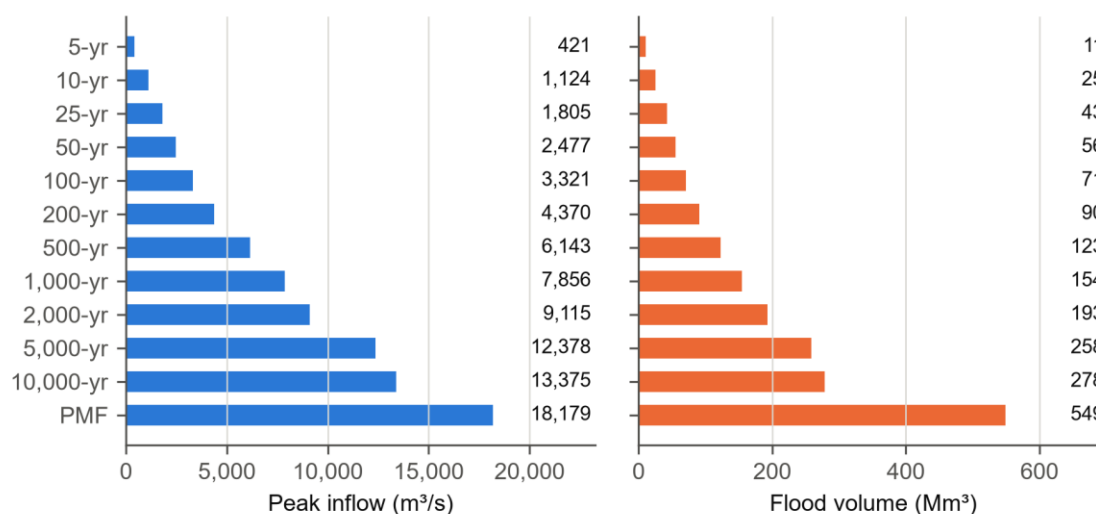


Figure 10 - Peak inflow and flood volume at the Majlas dam by return period.



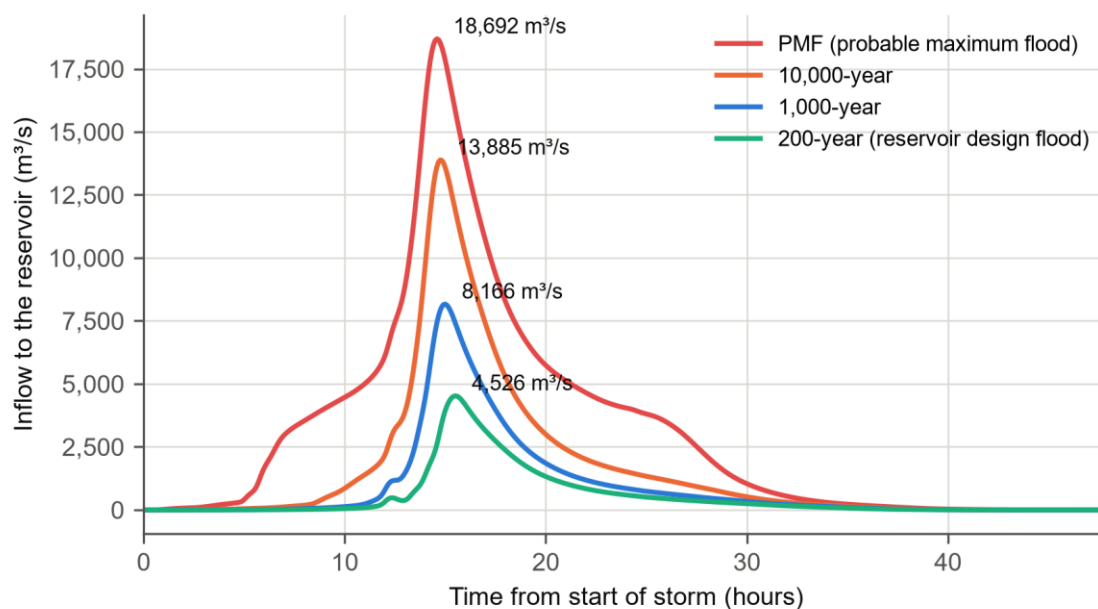


Figure 11 - Inflow hydrographs to the Majlas reservoir for the 200-, 1,000- and 10,000-year floods and the PMF (hydrology report).

2.6. DOWNSTREAM: QURAYAT AND THE PROTECTION WORKS

Below the dam the wadi leaves the gorge, crosses about 2 km of open floodplain and enters Qurayat, where it is confined by 2,600 m of training dikes and bank protection sized for the 200-year flood (Figure 12). The town holds 2,598 residential, 265 commercial, 1,043 agricultural and 5 industrial plots in the flood-risk study land-use layer, with the coast road, the wadi crossings and the utilities of the town in the path of the wadi. Figure 13 shows the plots and the 2025 population grid.

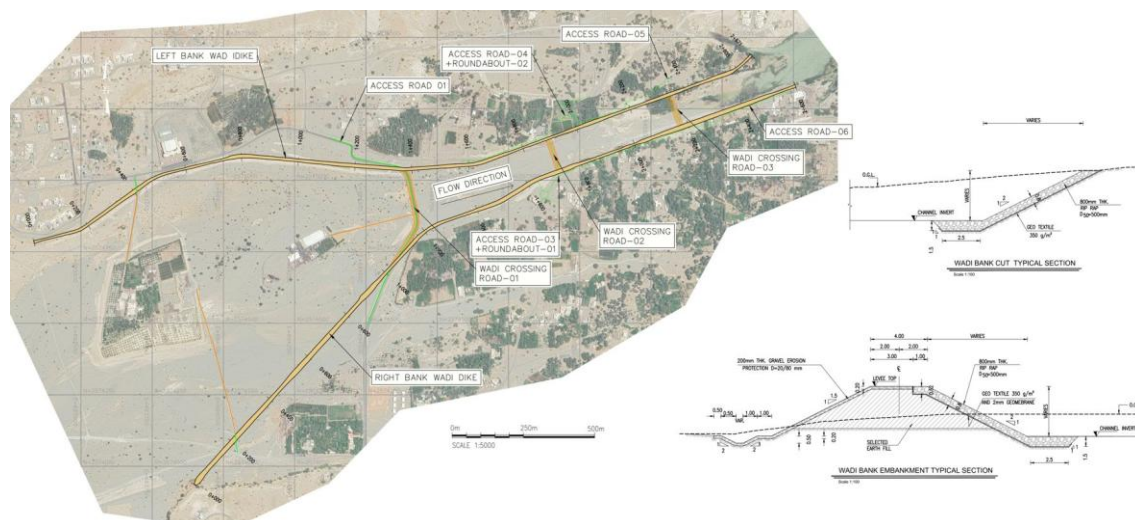


Figure 12 - Downstream protection works: training dikes, bank protection and wadi crossings (design review, September 2026).

Qurayat below the dam: use of each plot, population grid and protection works

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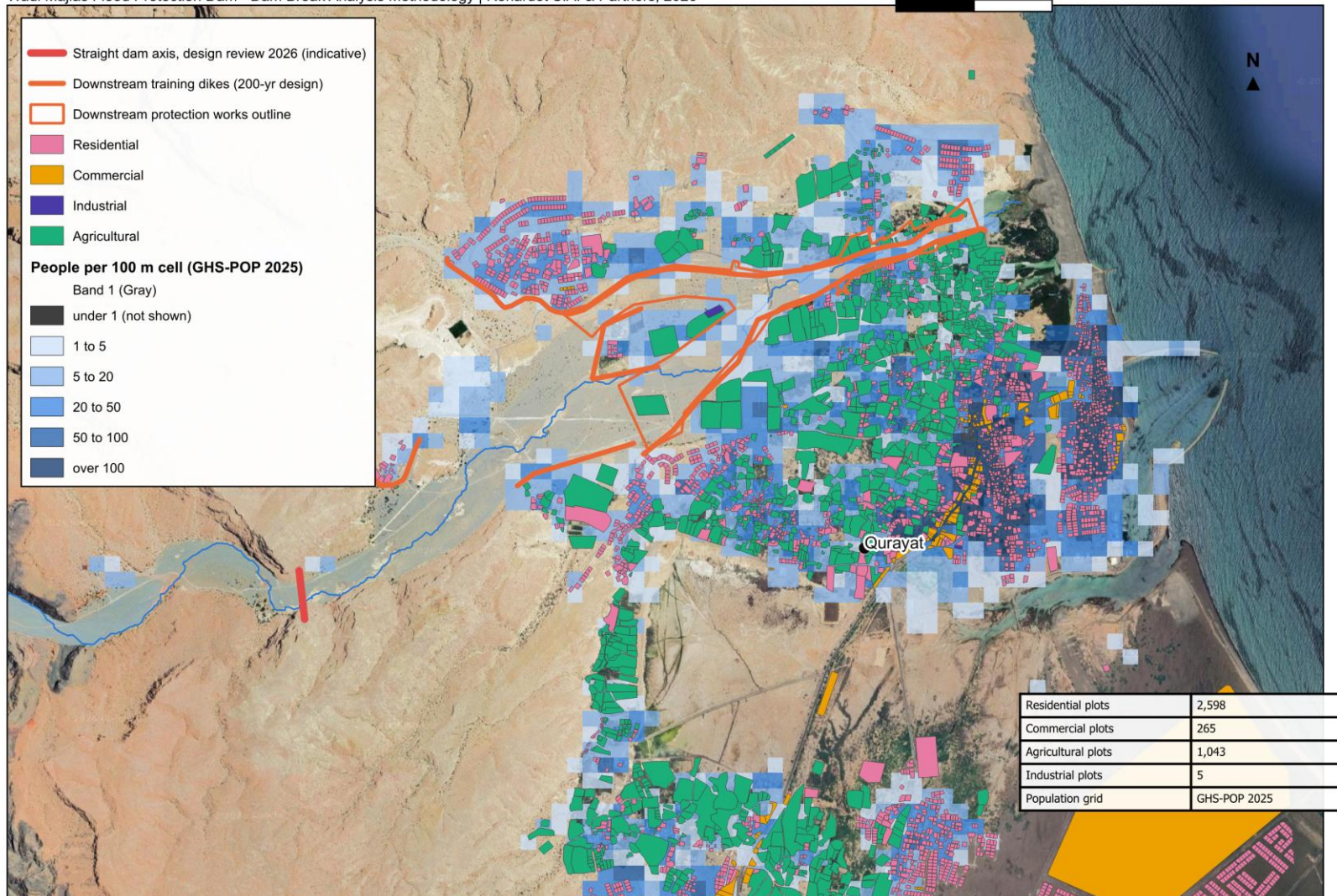


Figure 13 - Qurayat below the dam: use of each plot, the 2025 population grid and the downstream protection works.

2.7. DATA IN HAND

The study starts from a substantial base of data and models produced for the hydrology, the flood protection design and the flood-risk study (Table 8). The existing HEC-RAS 2D model of the downstream reach, with its terrain, land cover and dikes, is the starting point of the dam break model (Figure 14), and the terrain coverage is shown in Figure 15.

Table 8 - Data and models available at the start of the analysis.

Dataset	Source and resolution	Use in this study
National terrain model	NSA DSM, 2 m and 5 m grids, whole catchment and coast	Base terrain of the reservoir and downstream areas
Dam site survey	Design review: 2 m DTM of the dam and reservoir, 5 cm at the dam site	Priority terrain at the dam and the gorge (files to be supplied)
Design surfaces	Design review CAD: dam, spillway, stilling basin, dikes, roads	Burnt into the terrain (files to be supplied)
Existing HEC-RAS models	HEC-RAS 6.6 (geometry and most plans) and 7.0 (latest plan); downstream 2D area of 15.7 km ² and 12,680 cells; catchment-wide 2D area of 145,491 cells	Downstream mesh, land cover and dikes reused; mesh refined near the dam
Design floods and hydrographs	Hydrology report and HEC-HMS model, 5-minute step	Inflow to the reservoir for the flood-day and no-failure runs
Reservoir curves	Hydrology report, original axis	Check of the 2D reservoir volume; to be replaced by the final-axis curve
Land use	Flood-risk study cadastral plots with use class	Damage and asset counts
Population	GHS-POP 2025, 100 m cells	Population at risk and loss of life
Depth-damage curves	Flood-risk study, Oman-calibrated JRC functions	Economic damage
Flood hazard method	AIDR H1 to H6, as in the flood-risk study	Hazard maps



Study reach: existing HEC-RAS 2D area and proposed dam-break model extent

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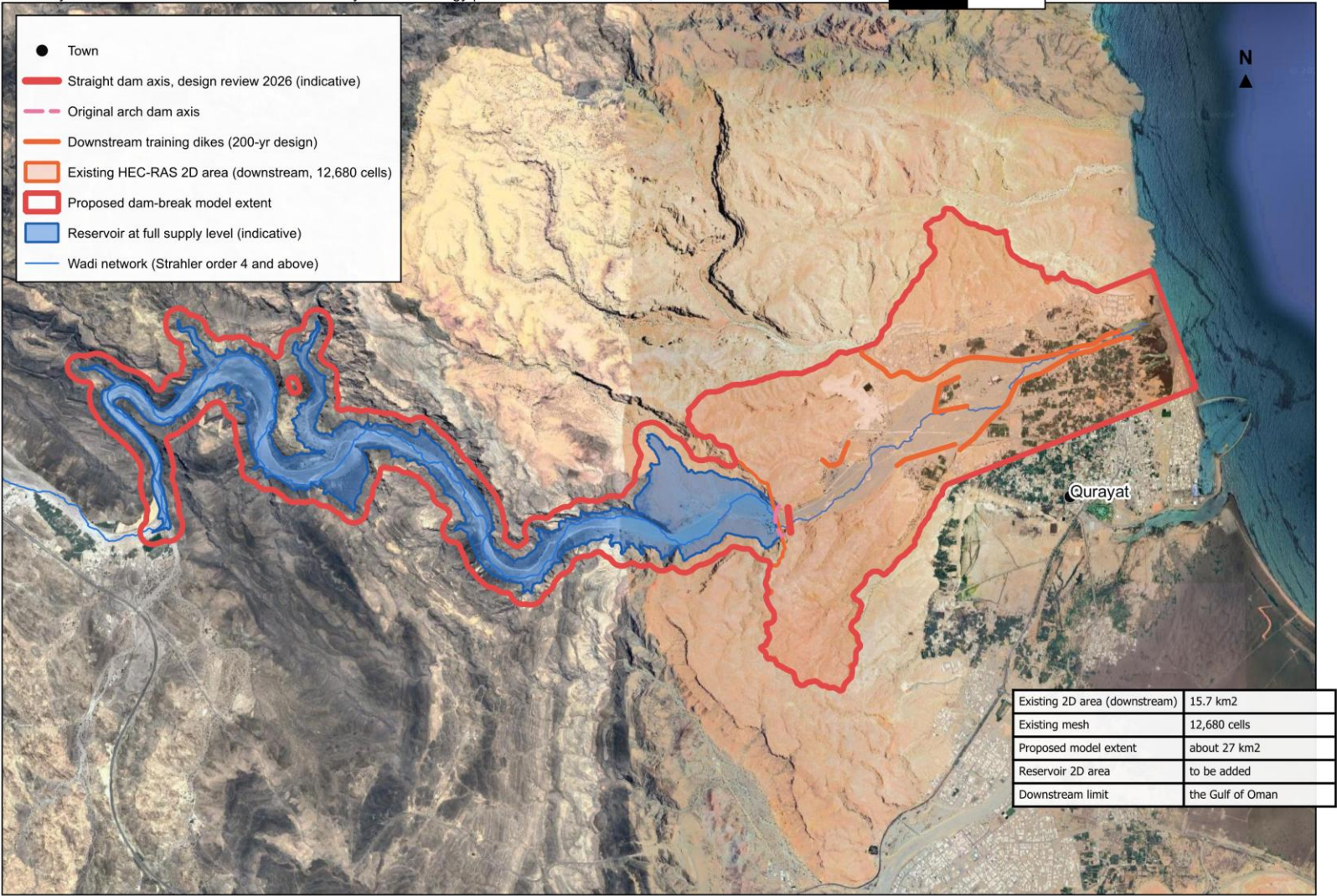


Figure 14 - Study reach: the existing HEC-RAS 2D area of the downstream reach and the proposed dam break model extent from the reservoir to the sea.





Terrain in hand: NSA 5 m terrain of the existing downstream model

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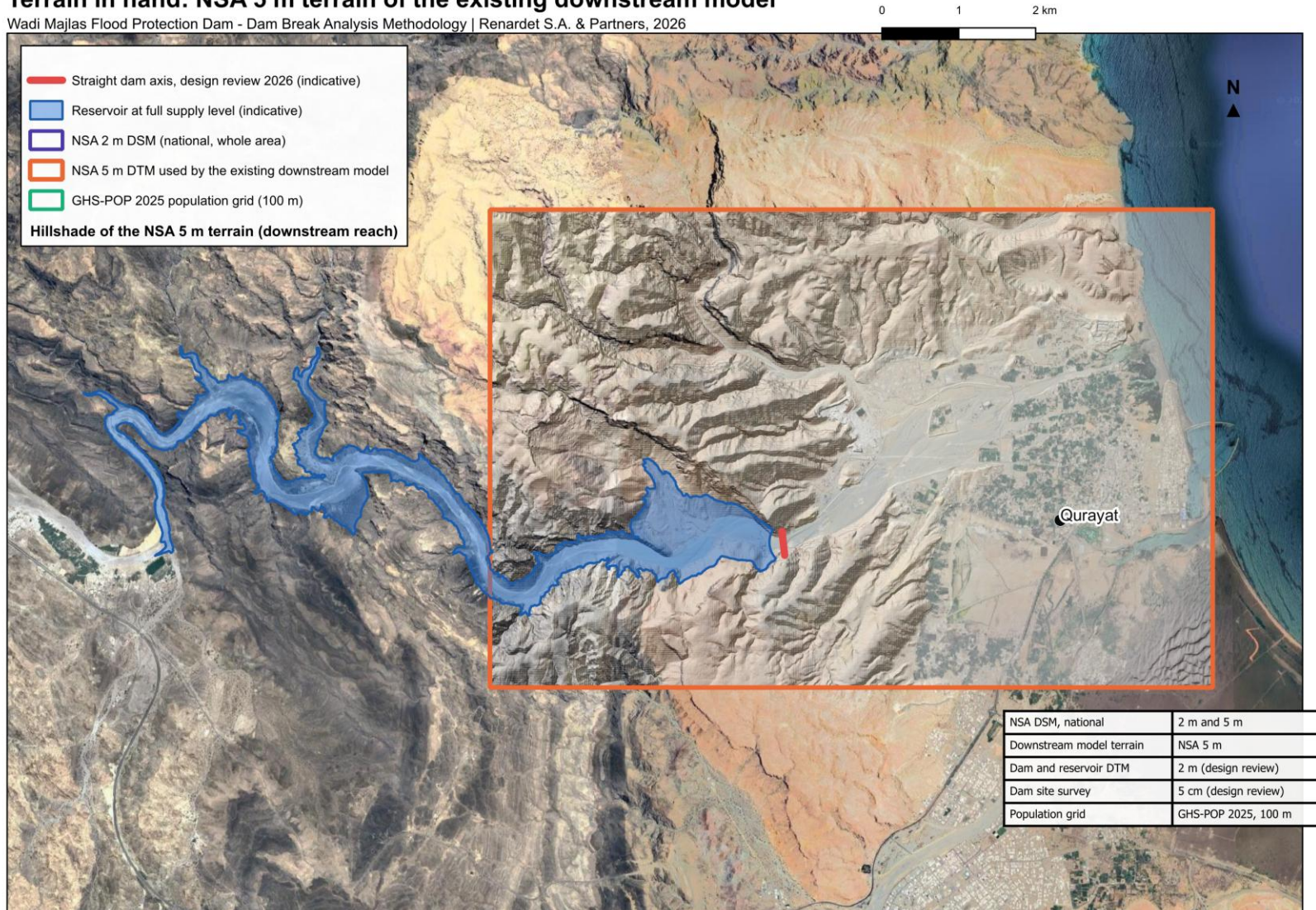


Figure 15 - Terrain in hand: hillshade and extent of the NSA 5 m terrain of the existing downstream model; the 2 m national DSM covers the whole area.

3. STANDARDS AND GUIDANCE FOLLOWED

3.1. POSITION IN OMAN

The Consultant is not aware of a published Omani standard that prescribes how a dam break analysis is to be carried out. The Client's requirements therefore rank first, and where they are silent this methodology follows the international guidance listed in Table 9, in the order given. The choice favours documents that are public, current, and written for regulators rather than for one software.

3.2. DOCUMENTS AND THEIR USE

Table 9 - Standards and guidance followed, in order of precedence.

Rank	Document	Used for
1	Client requirements and the scope of services	Scenarios to report, deliverables, map formats
2	FEMA P-946 (2013), Federal Guidelines for Inundation Mapping of Flood Risks Associated with Dam Incidents and Failures	Overall procedure, scenarios, map content, incremental approach
3	USACE HEC, Using HEC-RAS for Dam Break Studies, TD-39 (Brunner, 2014) and the HEC-RAS 6.x reference and user manuals	Model set-up, breach parameters for concrete dams, breach coefficients, checks
4	FERC Engineering Guidelines, Chapter 2 (1993) and USACE guidance as tabulated in the HEC-RAS reference	Breach width, side slopes and formation time for concrete gravity dams
5	USBR RCEM (2015), Reclamation Consequence Estimating Methodology, and USBR/USACE Best Practices in Dam and Levee Safety Risk Analysis (2019)	Loss-of-life estimate; failure modes of concrete gravity dams
6	ANCOLD (2012), Guidelines on the Consequence Categories for Dams	Consequence category of the dam
7	AIDR Guideline 7-3 (2017) and Smith, Davey and Cox (2014)	Flood hazard classes H1 to H6
8	ICOLD Bulletin 111 (1998), Dam-Break Flood Analysis	General principles and review of methods
9	Froehlich (2008), Wahl (1998, 2004), Xu and Zhang (2009), Von Thun and Gillette (1990), MacDonald and Langridge-Monopolis (1984)	Embankment breach equations, reported for reference only

3.3. WHAT THE INTERNATIONAL PRACTICE HAS IN COMMON

The documents differ in detail but agree on five points, and this methodology is built on them.

- The failure scenarios cover a failure with no flood (sunny day) and a failure during the inflow design flood (flood day), and the design flood is also run without failure so that the effect of the dam failing can be separated from the effect of the flood.
- The breach is described by its shape, size and formation time, chosen for the dam type, and the uncertainty in these values is shown by sensitivity runs rather than by a single conservative guess.
- The flood wave is routed with an unsteady hydraulic model; for a wide floodplain and a town in the path, a 2D model with the full flow equations.



- The results are maps of depth, velocity, arrival time and hazard at a scale useful for emergency planning, plus the population and assets in the flooded area.
- The consequences, and in particular the incremental loss of life, decide the hazard or consequence category of the dam, which in turn sets the design flood, the surveillance and the emergency planning requirements.



4. KINDS OF DAMS AND HOW EACH ONE BREAKS

4.1. WHY THE DAM TYPE COMES FIRST

The size of a dam break flood depends more on how fast the opening forms and how wide it is than on anything else in the analysis. Those two things are set by what the dam is made of. This chapter describes the main kinds of dams, how each kind fails, and how its breach is represented in a model, and then states which rules apply to Majlas. Figure 16 summarises the chapter.



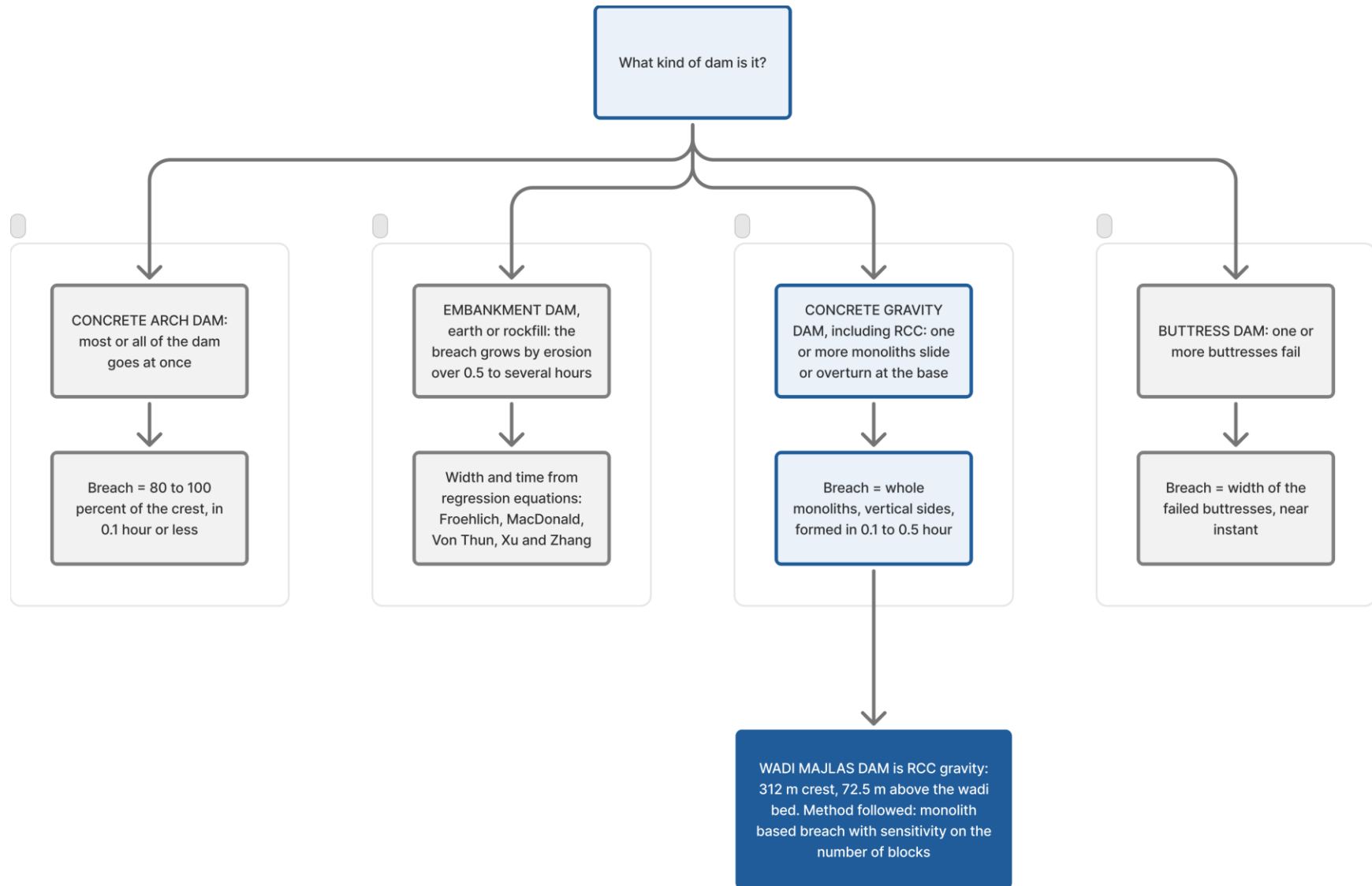


Figure 16 - How the dam type decides the way the breach is modelled, and the path that applies to the Wadi Majlas dam.

4.2. EMBANKMENT DAMS

Earthfill and rockfill dams are built of compacted soil or rock, made watertight by a clay core, a concrete or asphalt face, or a geomembrane. They fail by erosion. Either water flows over the crest and eats into the downstream face until a channel cuts through (overtopping), or water finds a path through or under the dam, carries soil with it, and the path grows into a tunnel that collapses (piping)¹. In both cases the opening grows over tens of minutes to several hours, its sides slope, and its final width depends on the height of the dam and the volume of water behind it. Because so many embankments have failed, there are statistical relations between the dam and reservoir size and the breach width and time. These are the regression equations² of Froehlich, MacDonald and Langridge-Monopolis, Von Thun and Gillette, and Xu and Zhang, and they are the standard tool for embankment dams. The earlier Al Dreez study used them correctly for a geomembrane-faced rockfill dam.

4.3. CONCRETE GRAVITY DAMS

A concrete gravity dam holds the water back by its own weight. It is built as a row of separate blocks, called monoliths³, each standing on the foundation. Concrete does not erode in the time a flood lasts, so a gravity dam does not breach by erosion and it can be overtopped without failing, provided the foundation at the toe is not washed out. It fails when the forces on a block exceed what the foundation can hold: the block slides forward on its base or on a weak layer in the rock, or it overturns. This can be triggered by an earthquake, by excessive water pressure under the base if the drains are blocked, by a weak seam or fault in the foundation, or by scour at the toe during a large

¹ Piping is internal erosion: seepage water washes fine particles out of the soil along a path, the path widens into a pipe, and the material above it falls in. It is the second most common cause of embankment failure after overtopping.

² A regression equation is a formula fitted to a set of recorded cases so that it reproduces them on average. Froehlich (2008), for example, fitted 74 embankment failures. Such an equation is only valid for dams like those in the data set.

³ A monolith is one block of the dam between two vertical contraction joints. A concrete dam is cast as a row of such blocks, typically 15 to 25 m wide, so that the concrete can shrink and move a little without cracking. Waterstops in the joints keep the reservoir water out.



flood. Once a block moves it leaves the dam almost at once, and its neighbours may follow. The opening is therefore rectangular with vertical sides, its width is a whole number of monoliths, and it forms in minutes. The United States federal guidance gives a breach width that is usually not more than half the crest length, vertical sides, and a formation time of 0.1 to 0.5 hour (USACE) or 0.1 to 0.3 hour (FERC). The regression equations of the previous section do not apply, and the HEC-RAS reference says so explicitly. Roller-compacted concrete dams are gravity dams in every respect that matters here; the placement method does not change how they fail.

4.4. CONCRETE ARCH DAMS

An arch dam is a thin curved wall that carries the water load sideways into the valley walls. If an abutment gives way the whole arch loses its support and most or all of the dam goes at once, as at Malpasset in 1959. The guidance therefore takes the breach as 80 to 100 percent of the crest length, with sides following the valley walls, formed in 0.1 hour or less. The original Majlas design was an arch; had it been kept, the breach would have been taken as the whole dam.

4.5. BUTTRESS AND COMPOSITE DAMS

A buttress dam is a sloping slab or a row of arches held up by concrete buttresses. It fails when one or more buttresses fail, and the breach is the width of the failed buttresses, near instant. Composite dams, with a concrete section in the river and embankment wings, are analysed section by section: the concrete part with the gravity or arch rules and the wings with the embankment rules, and the worse case governs.

4.6. SUMMARY OF THE RULES BY DAM TYPE

Table 10 - Breach rules by dam type.

Dam type	How it fails	Breach width	Side slopes	Formation time	Source
Earthfill / rockfill	Erosion by overtopping or piping	From regression equations; typically 1 to 5 times the height	0.5 to 1.4 H:1V	0.5 to several hours	Froehlich 2008, Wahl 1998, HEC-RAS reference
Concrete gravity (including RCC)	Monoliths slide or overturn on the foundation	Whole monoliths; usually not more than 0.5 of the crest length	Vertical	0.1 to 0.5 h (USACE), 0.1 to 0.3 h (FERC)	FERC 1993, USACE 1980 and 2007, HEC-RAS reference
Concrete arch	Abutment or foundation failure; whole arch goes	0.8 to 1.0 of the crest length	Valley wall slope	0.1 h or less	FERC 1993, USACE, HEC-RAS reference
Buttress	Buttresses fail	Width of the failed buttresses	Vertical	Near instant	FERC 1993



4.7. WHICH RULES APPLY TO MAJLAS

The Majlas dam is an RCC gravity dam. The concrete gravity rules apply: the breach is one or more whole monoliths with vertical sides, formed in about 0.1 hour, and its width is set from the block layout, not from an equation. The grout-enriched facing and the alluvium under the central blocks do not change the class of the dam; they change which blocks are most likely to move, which is why the failure is placed at the deepest section. Chapter 6 sets the values.

4.8. FAILURE MODES SCREENED FOR MAJLAS

Before choosing scenarios, the ways in which this particular dam could release its reservoir were listed and screened (Table 11). The screening decides which physical mechanism each scenario represents; it does not rate the likelihood of any of them.

Table 11 - Failure modes screened for the Majlas dam.

Mode	Credible for Majlas?	Why	Covered by
Sliding of monoliths on the foundation, normal pool	Yes	Deep alluvium and a fault-controlled limestone foundation; an earthquake or a loss of drainage could trigger it	Sunny-day scenario
Sliding or overturning of monoliths at the PMF pool	Yes	Highest water load and uplift; the pool stands about 1.2 m above the crest, held by the parapet	Flood-day scenario
Scour of the foundation at the toe during the PMF	Yes, as a trigger	Unit discharge of 85 m ² /s over the stepped chute; the stilling basin protects the toe but the abutment contact is the weak point	Flood-day scenario (same breach)
Overtopping erosion of the dam body	No	RCC does not erode during a flood; gravity dams are overtoppable	-
Piping through the dam body	No	There is no soil in the body; the facing is watertight concrete	-
Internal erosion of the alluvium under the cut-off	As a precursor only	It would weaken the foundation and lead to sliding, which is the sunny-day case	Sunny-day scenario
Spillway gate failure	Not applicable	The spillway is ungated	-
Bottom outlet failure	No dam failure	A gate or pipe failure releases at most 120 m ³ /s	-
Cascade failure of the Haifaz dam	Excluded	Small catchment; the Majlas reservoir absorbs the volume below the spillway crest	-



5. FAILURE SCENARIOS AND LOAD CASES

5.1. THE THREE SCENARIOS

Each scenario is a combination of a reservoir level, an inflow and a decision on whether the dam fails. Three are analysed.

- **Sunny day (S1).** The reservoir stands at the full supply level of 84.5 m with no flood in the wadi¹. The dam fails and the reservoir empties through the breach. This scenario gives the shortest warning time and the flood that is entirely due to the dam.
- **Flood day (S2).** The PMF enters the reservoir, the spillway runs at full capacity, the pool reaches 96.7 m and the dam fails at that moment². This scenario gives the highest flood levels downstream.
- **PMF without failure (S3).** The same flood passes the dam through the spillway and over the crest, and the dam holds. This is not a failure scenario; it is the baseline that shows what the flood would do anyway. The difference between S2 and S3 is the incremental effect of the failure³, which decides the classification.

5.2. WHY THE FLOOD-DAY CASE USES THE PROBABLE MAXIMUM FLOOD

The reservoir stores every flood up to the 200-year event below the spillway crest, so smaller floods cannot load the dam beyond its normal pool. The PMF is the inflow design flood of a dam whose failure could cause loss of life, and its routed pool of 96.7 m gives the largest water load and the highest starting level for a breach. Running the flood-day case with a smaller flood would understate the consequences and would not match the design basis of the spillway. The 10,000-year flood will

¹ The sunny-day scenario is the failure that comes without warning from the weather: on a dry day, for example after an earthquake. It gives the shortest warning time and, because the wadi downstream is dry, the clearest picture of the flood caused by the dam alone.

² The flood-day scenario, also called the rainy-day or hydrologic scenario, is a failure during the largest flood. The flood wave from the breach adds to a wadi that is already in flood, so the water levels downstream are the highest of all scenarios.

³ The incremental effect is what the dam failure adds to the flooding that would have happened without it. Dam classification in FEMA and ANCOLD practice rests on this increment, because a dam is not blamed for a flood it did not cause.



be run as a sensitivity case to show how much of the flood-day result depends on the choice of the PMF.

5.3. WHEN THE DAM FAILS

In the sunny-day scenario the failure is triggered by time: the model runs with a steady pool at 84.5 m for a warm-up period of about two hours to let the reservoir settle, and the breach starts at the end of it. In the flood-day scenario the failure is triggered by the pool level: the breach starts when the reservoir first reaches 96.7 m, which is close to the peak of the inflow. A failure at the exact peak is the standard assumption because it maximises the outflow; the sensitivity runs include a failure two hours after the peak, when the pool is still high but the inflow is falling, to show the effect.

5.4. DOWNSTREAM CONDITION: WITH AND WITHOUT THE DIKES

The training dikes through Qurayat are designed for the 200-year flood and will be overwhelmed by a breach wave many times larger. They still matter: for the first minutes they hold the water in the channel and change where it first spills, and they may fail when overtopped. Every scenario is run twice, with the dikes in the terrain and without them, and both results are reported. The with-dikes case represents the scheme as built; the without-dikes case represents the condition before construction and the case of the dikes failing early. The dikes are not breached in the model; the difference between the two runs brackets the effect.

5.5. OTHER CONDITIONS FIXED FOR ALL RUNS

- The bottom outlet is closed and the groundwater conveyance pipes are ignored; their discharge is negligible against the breach flow.
- The downstream boundary at the Gulf of Oman is held at the highest astronomical tide for Qurayat, to be taken from the national tide tables; a mean sea level case is run as a sensitivity.
- The wadi downstream is dry at the start of the sunny-day run, and carries the local runoff of the PMF storm in the flood-day and no-failure runs, taken from the HEC-HMS model at the junctions below the dam.
- The reservoir is at its design geometry; sedimentation over the life of the dam is not included, because it reduces the stored volume and would reduce the breach flood.

5.6. THE RUNS

Table 12 lists the six base runs. The sensitivity runs are defined in Chapter 10.



Table 12 - Base runs of the dam break analysis.

Run	Scenario	Starting pool (m a.s.l.)	Inflow	Dam	Dikes	Purpose
S1-D	Sunny day	84.5	None	Fails at end of warm-up	In place	Dam-only flood, shortest warning
S1-N	Sunny day	84.5	None	Fails at end of warm-up	Removed	Same, before the dikes are built or after they fail
S2-D	Flood day	96.7 at failure	PMF	Fails when the pool reaches 96.7 m	In place	Highest flood levels
S2-N	Flood day	96.7 at failure	PMF	Same	Removed	Same, without the dikes
S3-D	PMF, no failure	Rises with the flood	PMF	Holds; spillway and crest overflow	In place	Baseline for the incremental effect; routing check
S3-N	PMF, no failure	Rises with the flood	PMF	Holds	Removed	Baseline without the dikes



6. BREACH PARAMETERS FOR THE RCC GRAVITY DAM

6.1. WHAT THE MODEL NEEDS

HEC-RAS describes a breach with a short list of numbers. Each is explained below, because the choice of each one is a decision of this methodology and not a software default.

- **Location.** The station along the dam axis at which the opening is centred.
- **Bottom elevation.** The level of the floor of the opening once it is fully formed.
- **Bottom width.** The width of the opening at its floor.
- **Side slopes.** The slope of the two sides of the opening, given as horizontal to vertical; zero means vertical sides.
- **Formation time.** The time from the start of the opening to its full size¹.
- **Progression.** How the opening grows during the formation time: at a constant rate (linear) or slowly at first and fast in the middle (sine curve).
- **Trigger.** What starts the breach: a clock time, a pool elevation, or a pool elevation held for a duration.
- **Weir coefficient.** The coefficient in the weir equation that converts the head over the breach floor into a discharge².

6.2. HOW EACH PARAMETER IS SET FOR MAJLAS

Location. At the deepest part of the dam, which is the spillway section over the wadi bed. The blocks there carry the highest water load, stand on the alluvium-filled channel, and their loss releases water from the lowest level. Failure of an abutment block would release less water from a higher floor and is bounded by the central case.

Bottom elevation. The wadi bed at the axis, 23.0 m a.s.l. The foundation is excavated to -15.0 m, but the failed blocks slide onto ground that stands at 23 to 30 m, and the stilling basin and rockfill downstream remain in place, so the water cannot leave below the natural bed. Taking the bed rather than the foundation level is the standard practice and is slightly conservative for the peak, because a lower floor would drown the outflow in its own tailwater.

¹ In HEC-RAS the formation time starts when water begins to leave through the opening and ends when the opening has reached its full size. It is not the time to empty the reservoir, which the model computes.

² Flow through the breach is computed as flow over a broad-crested weir, $Q = C B H^{1.5}$, with C the weir coefficient in $m^{0.5}/s$, B the width and H the head above the breach floor. HEC-RAS quotes C in US units; the values in this report are converted to metric.



Width. A whole number of blocks. The spillway bays are 40 m wide between 3 m piers, so one bay with its pier is 43 m. The base case takes two bays, 86 m, which is 0.28 of the crest length and within the federal ceiling of half the crest. The lower case takes one bay, 43 m, and the upper case half the crest length, 156 m, which is the ceiling itself. A three-bay case of 129 m is run to show the trend. When the design team supplies the joint layout the widths will be rounded to the actual blocks.

Side slopes. Vertical. The sides of the opening are the contraction joints of the blocks that remain.

Formation time. 0.1 hour, 6 minutes, in the base case, with 0.05 and 0.3 hour as the sensitivity range. The federal guidance gives 0.1 to 0.5 hour (USACE) and 0.1 to 0.3 hour (FERC) for gravity dams; the short end is used because a sliding block clears the dam in the time the water needs to push it, and a longer time only lowers the peak.

Progression. Linear. The sine curve models the slow start of an erosion breach and is not appropriate for a block that moves as a unit (Figure 17).

Trigger. Clock time at the end of the warm-up in the sunny-day runs; pool elevation of 96.7 m in the flood-day runs.

Weir coefficient. 1.44 to 1.66 $\text{m}^{0.5}/\text{s}$, the HEC-RAS range for concrete gravity dams (2.6 to 3.0 in US units); the upper value is used in the base case because the opening has a sharp, clean edge.

Table 13 collects the values and Table 14 the guidance they come from.

Table 13 - Breach parameters for the Majlas dam.

Parameter	Base case	Lower case	Upper case	Basis
Location	Centre of the spillway section	Same	Same	Deepest blocks, highest load
Bottom elevation	23.0 m a.s.l.	Same	Same	Wadi bed at the axis
Bottom width	86 m (two bays)	43 m (one bay)	156 m (half the crest)	Block layout; federal ceiling of 0.5 L
Side slopes	Vertical (0 H:1V)	Same	Same	Contraction joints
Formation time	0.1 h	0.3 h	0.05 h	USACE 0.1 to 0.5 h; FERC 0.1 to 0.3 h
Progression	Linear	Same	Same	Block moves as a unit
Trigger, sunny day	Clock time, end of 2 h warm-up	Same	Same	Steady pool



Parameter	Base case	Lower case	Upper case	Basis
Trigger, flood day	Pool at 96.7 m	Same	Pool at peak, plus a run 2 h after the peak	Highest load
Weir coefficient	$1.66 \text{ m}^{0.5}/\text{s}$	$1.44 \text{ m}^{0.5}/\text{s}$	$1.66 \text{ m}^{0.5}/\text{s}$	HEC-RAS range for gravity dams

Table 14 - Federal agency guidance on breach parameters for concrete dams (HEC-RAS 1D Technical Reference, Estimating breach parameters; L = crest length).

Dam type	Average breach width	Side slopes	Failure time (h)	Agency
Concrete gravity	Usually not more than 0.5 L , multiple monoliths	Vertical	0.1 to 0.5	USACE 1980 and 2007
Concrete gravity	Usually not more than 0.5 L , multiple monoliths	Vertical	0.1 to 0.3	FERC
Concrete gravity	Usually not more than 0.5 L , multiple monoliths	Vertical	0.1 to 0.2	NWS
Concrete arch	0.8 L to L , or the entire dam	0 to the valley walls	0.1 or less	USACE, FERC, NWS

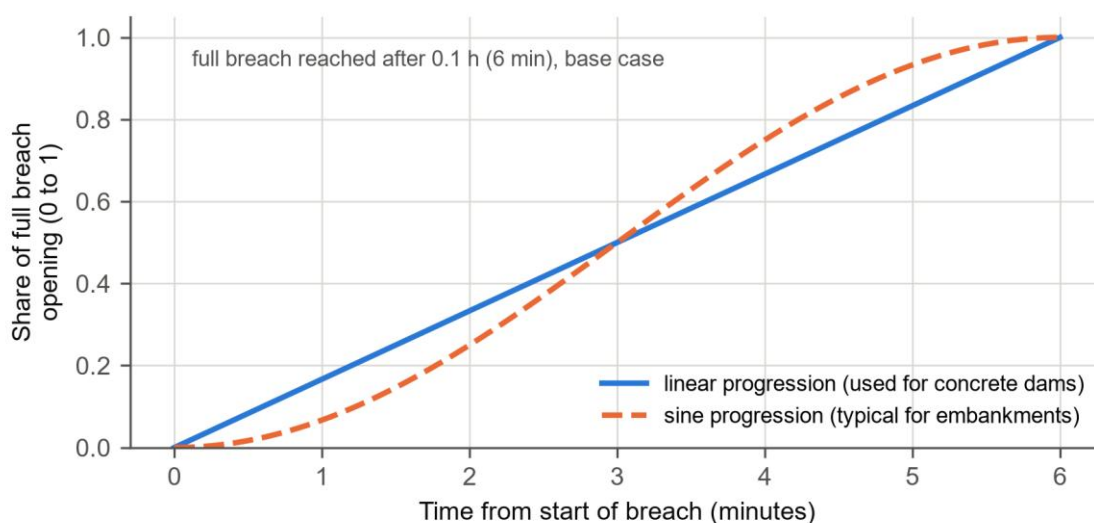


Figure 17 - Growth of the opening during the formation time: linear progression, used here, against the sine progression used for erosion breaches (base case 0.1 h).



6.3. HAND CHECK OF THE PEAK OUTFLOW

Before any model is run, the order of magnitude of the peak outflow is fixed by two hand calculations, and the model result must fall between them or be explained. The first is the weir equation applied to the full opening at the starting pool, before the reservoir has dropped:

$$Q_p = CB(H - Z_b)^{1.5} \quad (1)$$

where:

Q_p peak outflow through the breach (m^3/s)

C weir coefficient, $1.7 \text{ m}^{0.5}/\text{s}$ for the check

B breach width (m)

H starting pool level (m a.s.l.)

Z_b breach bottom, 23.0 m a.s.l.

The second is the classical solution for the sudden removal of a wall from a reservoir onto a dry bed (Ritter, 1892), which gives a lower value because it accounts for the drop of the water surface at the opening:

$$Q_p = \frac{8}{27} B \sqrt{g} (H - Z_b)^{1.5} \quad (2)$$

where:

g gravity, 9.81 m/s^2

B, H, Z_b as in equation (1)

Figure 18 shows both for the four widths and the two pools. For the base case of 86 m the first-minute peak lies between about 38,000 and 71,000 m^3/s on a sunny day and between 50,000 and 93,000 m^3/s on a flood day. These are check values only: the modelled peak will be lower because the pool



drops during the six minutes of formation and the tailwater rises in the stilling basin, and it will be reported against these bounds in the analysis report.

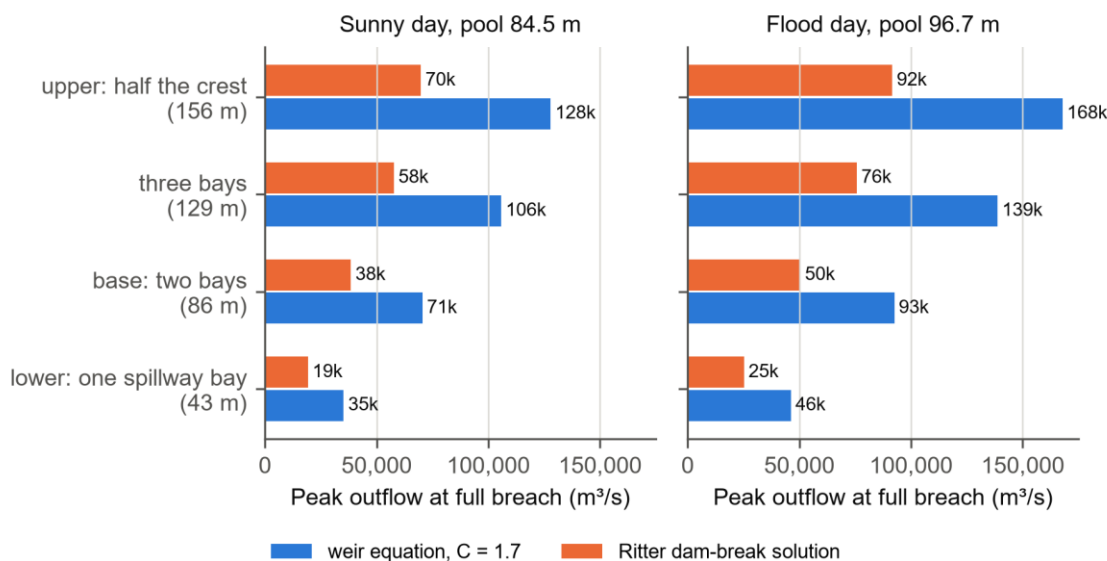


Figure 18 - Hand check of the peak outflow the moment the breach is fully open, for the four breach widths and the two starting pools; weir equation and Ritter solution.

6.4. WHY THE REGRESSION EQUATIONS ARE NOT USED

The Froehlich, MacDonald, Von Thun and Xu and Zhang equations are reported in Appendix A for completeness, with the values they would give for Majlas. They are not used, for three reasons. They were fitted to embankment failures, and the HEC-RAS reference states that they do not apply to concrete dams. They give formation times of one to several hours, which for a concrete dam would understate the peak by a large factor and overstate the warning time. And their sloping sides have no physical meaning between two contraction joints. Their only use here is to show the reader how different the answer would have been.



7. HYDRAULIC MODEL

7.1. SOFTWARE AND TYPE OF MODEL

The flood wave is computed with HEC-RAS, version 6.6, in two dimensions. The existing models of the scheme are in the same version, the software is free and documented, and its dam break tools are the ones the international guidance is written around. A 2D model is required rather than a 1D model¹ because below the gorge the wave spreads over a plain several kilometres wide and through a town, and the maps must show where the water goes, not only how high it stands.

The model solves the full shallow water equations² in every failure run. The diffusion wave option is used only to warm up the reservoir before the breach and, if needed, to speed up the no-failure run once it has been checked against the full equations.

7.2. LAYOUT OF THE MODEL

The model has two 2D flow areas joined by one connection (Figure 19). The reservoir area covers the valley from the dam to the upstream end of the pool at the PMF level, on the 2 m terrain. The downstream area is the existing 15.7 km² area of the flood protection model, extended to the shoreline and refined near the dam. The connection is the dam: it carries the non-overflow crest, the spillway, the piers and the breach. Modelling the reservoir as a 2D area rather than as a storage area with a volume curve is the choice recommended for sudden failures, because it reproduces the drawdown wave that runs up the reservoir and limits the outflow in the first minutes; a storage-area run is kept as a sensitivity check. The catchment-wide model of the hydrology study is not used,

¹ A 1D model computes one water level and one velocity per cross-section across the whole valley. A 2D model divides the ground into cells and computes depth and velocity in each, so it can show water spreading over a floodplain, splitting around a town and returning to the wadi. For a dam break wave entering an open coastal plain, only the 2D model gives usable maps.

² The shallow water equations are the equations of mass and momentum for water whose depth is small compared with its extent. HEC-RAS offers a simplified form, the diffusion wave, that ignores the inertia of the water. The inertia is what carries a dam break wave, so the diffusion wave must not be used for the failure runs.



because its 5 m cells and its size make the failure runs too slow, and the inflow it would provide is already available as hydrographs.



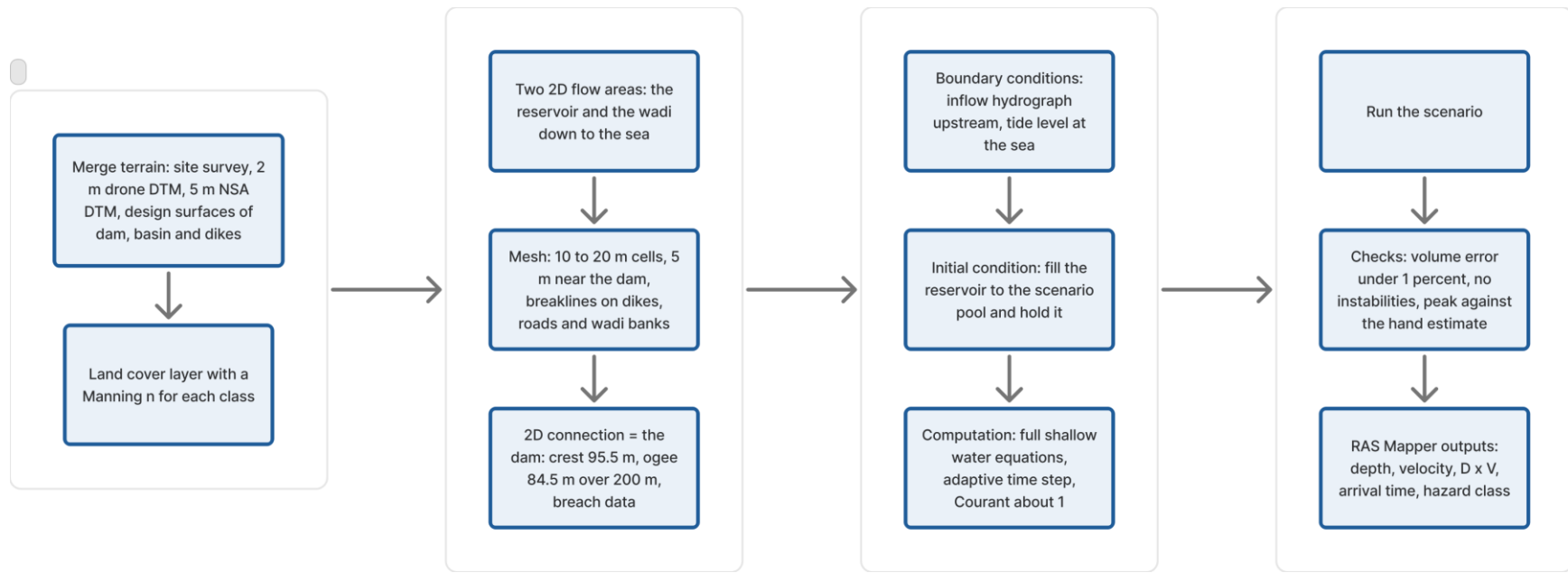


Figure 19 - Order in which the HEC-RAS 2D dam break model is prepared, built, run and checked.

7.3. TERRAIN

One terrain is built for the whole model from the sources in Table 15, with the more accurate source taking priority where two overlap. The design surfaces are burnt into the ground so that the dam, the stilling basin, the dikes, the wadi crossings and the reservoir bed are as designed, not as the survey found them. The dam itself is then cut out of the terrain along the connection, because the connection represents it. In the no-dikes runs the dike surfaces are left out.

Table 15 - Terrain sources and their priority in the merged terrain.

Priority	Source	Resolution	Covers	Status
1	Dam-site topographic survey	5 cm	Dam site and gorge	To be supplied by the design team
2	Design surfaces: dam, spillway, stilling basin, bottom outlet chute, dikes, roads, wadi bank cuts	CAD	Where designed	To be supplied as CAD or DTM
3	Drone survey of the dam and reservoir area	2 m	Dam and reservoir	To be supplied
4	NSA DTM of the downstream reach used in the existing model	5 m	Downstream 2D area	In hand
5	NSA national DSM	2 m / 5 m	Everything else, including the coast	In hand

7.4. LAND COVER AND ROUGHNESS

Roughness¹ is assigned by land-cover class from the flood-risk study land-use layer and the existing model, with the values in Table 16. The base values are those of the existing downstream model where they exist, so that the dam break results can be compared with the flood protection results on the same basis.

Table 16 - Manning's n by land-cover class.

Land cover	Manning's n, base	Range for sensitivity	Where
Wadi bed, gravel and cobbles	0.035	0.030 to 0.045	Active channel above and below the dam
Floodplain, bare or sparse scrub	0.040	0.035 to 0.050	Plain between the gorge and the town
Agriculture and date gardens	0.080	0.060 to 0.100	Plots classed agricultural

¹ Roughness is the resistance the ground offers to the flow, expressed by Manning's n. Higher n means slower and deeper water. It is the parameter with the widest accepted range, so it is always varied in the sensitivity runs.



Land cover	Manning's n, base	Range for sensitivity	Where
Residential and commercial blocks	0.120	0.080 to 0.200	Built plots; buildings not cut into the terrain
Roads and paved surfaces	0.018	0.016 to 0.022	Coast road, town roads
Concrete structures	0.015	0.013 to 0.017	Spillway chute, stilling basin, dikes with riprap 0.035
Reservoir bed, rock and gravel	0.035	0.030 to 0.045	Reservoir 2D area
Sea and lagoon	0.025	0.020 to 0.030	Shoreline strip

7.5. MESH

The mesh is built from the existing downstream mesh and a new reservoir mesh. Cell size follows the need: small where the flow changes fast, larger where it is uniform. Breaklines¹ are placed along the dikes, the wadi banks, the road embankments and the coast road. The dam connection line is itself a breakline.

Table 17 - Mesh cell sizes by zone.

Zone	Cell size	Reason
Reservoir, main body	30 to 50 m	Uniform, deep water; volume must match the curve within 2 percent
Reservoir, last 1 km to the dam	10 m	Drawdown wave and approach flow to the breach
Dam and stilling basin	5 m	Breach outflow, jet and hydraulic jump
Gorge and plain to the town	10 to 15 m	Wave front and spreading
Town and dikes	10 m, breaklines on all dikes and roads	Flow around blocks, along streets and over the dikes
Coastal strip and sea	20 to 30 m	Outflow to the sea

¹ A breakline is a line along which the model aligns the faces of its cells, so that a dike, a road embankment or a wadi bank is represented as a sharp edge instead of a ramp smeared over a cell. Water then stays behind it until it truly overtops.



7.6. THE DAM AS A 2D CONNECTION

The dam is entered as a connection between the two 2D areas¹. Its weir profile is the crest of the dam: 95.5 m along the non-overflow blocks, 84.5 m over the five spillway bays, and the pier tops between them. The ogee bays use the ogee weir option with the design head from the spillway design; the non-overflow crest uses the broad-crested option. Table 18 lists the entries. The bottom outlet is not entered, because it is closed.

Table 18 - Entries for the dam connection in HEC-RAS.

HEC-RAS entry	Value	How it is obtained
Connection alignment	Straight axis, 312 m, from the design drawings	Design team CAD
Weir profile, non-overflow blocks	95.5 m a.s.l.	Crest level; the parapet is not credited
Weir profile, spillway bays	84.5 m over 5 × 40 m	Ogee crest level and bay widths
Weir profile, piers	96.7 m over 4 × 3 m	Pier tops
Weir shape and coefficient, spillway	Ogee; design head from the spillway design; C about 2.1 to 2.2 m ^{0.5} /s	Spillway design; checked against the rating of Chapter 2
Weir shape and coefficient, crest	Broad crested; 1.44 to 1.66 m ^{0.5} /s	HEC-RAS range for gravity dams
Weir width	6 m	Crest width
Breach method	User entered data	Chapter 6
Breach centre station	Centre of the spillway section	Chapter 6
Final bottom width	86 m base; 43 and 156 m in sensitivity	Chapter 6
Final bottom elevation	23.0 m a.s.l.	Chapter 6
Side slopes	0 H:1V	Chapter 6
Breach weir coefficient	1.66 m ^{0.5} /s	Chapter 6
Formation time	0.1 h base; 0.05 and 0.3 h in sensitivity	Chapter 6
Failure mode	Overtopping (the option that opens from the top down); piping is not used	Chapter 6
Trigger	Set time (sunny day); water surface elevation (flood day)	Chapter 5
Breach progression	Linear	Chapter 6

¹ In HEC-RAS a connection, called an SA/2D area connection, is a line structure between two areas with a weir profile along it, optional culverts and gates, and a breach. Flow over the weir is computed from the head on each side at every time step, so the spillway, the crest and the breach all discharge according to the water levels the model computes.



7.7. BOUNDARY CONDITIONS

Table 19 - Boundary conditions of the model.

Boundary	Type	Sunny day	Flood day and no failure
Reservoir, upstream end	Flow hydrograph	None (steady pool)	PMF inflow hydrograph from HEC-HMS, 5-minute step
Downstream tributaries	Flow hydrographs at the HEC-HMS junctions below the dam	None	Local PMF runoff of the downstream sub-catchments
Sea	Stage hydrograph	Highest astronomical tide, constant	Same; mean sea level in sensitivity
Open land edges	Normal depth	Slope from the terrain	Same

7.8. INITIAL CONDITIONS

The reservoir starts full. The sunny-day runs set the initial water surface of the reservoir area to 84.5 m and hold it for a two-hour warm-up before the breach, which lets any small waves from the start of the computation die away. The flood-day and no-failure runs start at 84.5 m with the PMF hydrograph and let the pool rise with the flood; the failure is then triggered by the pool level. The downstream area starts dry except for the sea. A restart file is written at the end of the warm-up so that the sensitivity runs can start from the same state.

7.9. COMPUTATION SETTINGS

Table 20 gives the settings. The time step is adaptive and limited by the Courant number¹, so that the model keeps a short step while the wave front passes and a longer one afterwards:

$$Cr = \frac{V\Delta t}{\Delta x} \leq 1 \quad (3)$$

where:

¹ The Courant number is the distance a wave travels in one time step divided by the cell size. If it is above 1 the wave crosses more than one cell per step and the solution can become unstable or lose the peak. HEC-RAS can halve and double the time step to keep it in range.



Cr Courant number

V flow velocity plus wave celerity (m/s)

Δt time step (s)

Δx cell size in the direction of flow (m)

Table 20 - Computation settings for the failure runs.

Setting	Value	Reason
Equation set	Shallow water equations, Eulerian-Eulerian (SWE-EM)	Momentum conserving; recommended for rapidly varying flow
Time step	Adaptive; maximum Courant 1.0; halve above, double below 0.5; initial 1 s	Stability at the wave front
Implicit weighting (theta)	1.0	Stability for the breach
Water surface tolerance	0.003 m	Convergence
Volume tolerance	0.003 m	Convergence
Maximum iterations	20	Convergence at the breach
Turbulence	Conservative, mixing coefficients default	Not decisive; noted
Hydrograph output interval	10 s at the dam connection and at reference lines	Peak capture
Mapping output interval	1 minute for the first two hours after the breach, 5 minutes after	Arrival time maps
Run duration	Sunny day: warm-up plus 12 h; flood day and no failure: 60 h	Full passage of the wave to the sea

7.10. RESERVOIR ROUTING CHECK BEFORE THE FAILURE RUNS

The no-failure PMF run is made first and checked against the design. The spillway discharge of the model at the PMF pool must match the rating of the spillway design and the capacity of 17,000 m³/s within 5 percent, the peak pool must match the routed maximum water level given by the design team within 0.2 m, and the reservoir volume of the 2D area at full supply level must match the elevation-volume curve within 2 percent. The rating follows the ogee equation:

$$Q = CLH^{1.5} \quad (4)$$

where:



Q spillway discharge (m³/s)

C discharge coefficient of the ogee, 2.0 to 2.2 m^{0.5}/s near the design head

L net crest length, 200 m

H head above the crest (m)

Only when these three checks pass are the failure runs started. Any difference is resolved with the design team, not adjusted in the model.

7.11. STABILITY AND VOLUME CHECKS ON EVERY RUN

Every run is checked for numerical trouble before its results are used. The checks are listed in Chapter 10; the first of them is the volume balance, which HEC-RAS reports as the volume accounting error:

$$\varepsilon_V = \frac{(V_{in} - V_{out} - \Delta S)}{V_{in}} \times 100 \quad (5)$$

where:

ε_V volume error (percent), accepted if below 1

V_{in} volume that entered the model

V_{out} volume that left through the boundaries

ΔS change of volume stored in the model



8. INUNDATION MAPS AND HAZARD CLASSIFICATION

8.1. PRODUCTS

Each run produces the rasters in Table 21 at 2 m resolution, the resolution used in the flood-risk study, so that the two sets of maps can be overlaid. All are read from the HEC-RAS results in RAS Mapper and exported to the project GIS, where the maps are laid out in the same template as the other maps of the project (Appendix C).

Table 21 - Map and time-series products of each run.

Product	Definition	Use
Maximum depth	Highest water depth reached in each cell during the run	Inundation extent, damage
Maximum velocity	Highest velocity reached in each cell	Hazard, scour
Maximum depth × velocity	Highest value of the product in each cell	Hazard class, loss of life
Arrival time	Time from the start of the breach to a depth of 0.3 m in the cell	Warning time, evacuation
Time to peak	Time from the start of the breach to the maximum depth	Evacuation
Duration	Time during which the depth exceeds 0.3 m	Recovery, access
Hazard class	AIDR class H1 to H6 from depth and depth × velocity	Danger to people, vehicles and buildings
Incremental depth	Maximum depth of the failure run minus that of the no-failure run	Classification; area affected by the dam alone
Breach hydrograph	Outflow through the dam connection against time	Peak, volume, comparison with the hand check
Reference-line hydrographs	Flow and level at the gorge exit, the town entry, the coast road and the shoreline	Arrival and peak times at the places that matter

8.2. HAZARD CLASSES

The danger of flowing water to people, vehicles and buildings is classified with the six classes of the Australian guideline, which are defined by depth, velocity and their product (Figure 20, Table 22). The same classes were used in the flood-risk study of the scheme, so the dam break hazard maps can be read against the flood hazard maps already delivered. HEC-RAS computes the classes directly in RAS Mapper.

$$D \times V = \text{depth} \times \text{velocity} \quad (6)$$



where:

D depth (m)

V velocity (m/s)

$D \times V$ product (m^2/s), the measure of the force on a person or a wall

Table 22 - Flood hazard classes H1 to H6 (AIDR Guideline 7-3, 2017; Smith, Davey and Cox, 2014). A cell takes the highest class that any of the three limits gives.

Class	$D \times V$ limit (m^2/s)	Depth limit (m)	Velocity limit (m/s)	Meaning
H1	0.3	0.3	2.0	Generally safe for people, vehicles and buildings
H2	0.6	0.5	2.0	Unsafe for small vehicles
H3	0.6	1.2	2.0	Unsafe for vehicles, children and the elderly
H4	1.0	2.0	2.0	Unsafe for people and vehicles
H5	4.0	4.0	4.0	Unsafe for people and vehicles; buildings vulnerable to structural damage
H6	above 4.0	-	-	Unsafe for everything; buildings fail

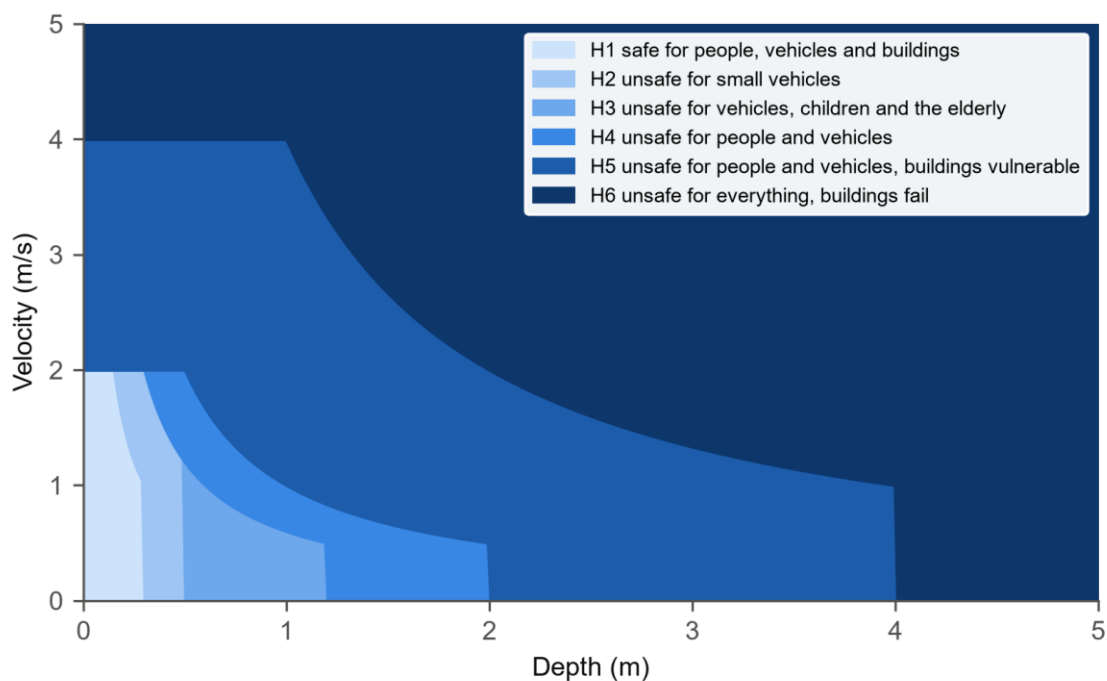


Figure 20 - Flood hazard classes H1 to H6 in the depth-velocity plane.

8.3. MAP SERIES

For each of the six base runs a general map of the whole reach at 1:60,000 and a map series of the town at 1:10,000 are produced for maximum depth, hazard class and arrival time, together with an incremental depth map for the flood-day run. Sensitivity runs are reported by their breach hydrographs and by summary maps of the envelope of the maximum depth. The maps carry the scenario, the run identifier, the breach parameters and the date of the model in their information box, so that a map can never be separated from the assumptions behind it.

8.4. INCREMENTAL MAPS

The incremental depth map is the flood-day result minus the no-failure result, cell by cell. Where the increment is below 0.6 m the dam failure adds little to the flood that would have occurred anyway; the 0.6 m threshold is the value the FEMA guidance uses to decide whether a failure changes the hazard to people at a location. The area above the threshold is the area for which the dam, rather than the flood, is responsible, and it is the area the classification looks at.



9. CONSEQUENCES AND DAM CLASSIFICATION

The maps say where the water goes. This chapter turns them into numbers that the Client can act on: how many people are in the way, how many might not survive, what the damage is, and what category the dam falls into (Figure 21).



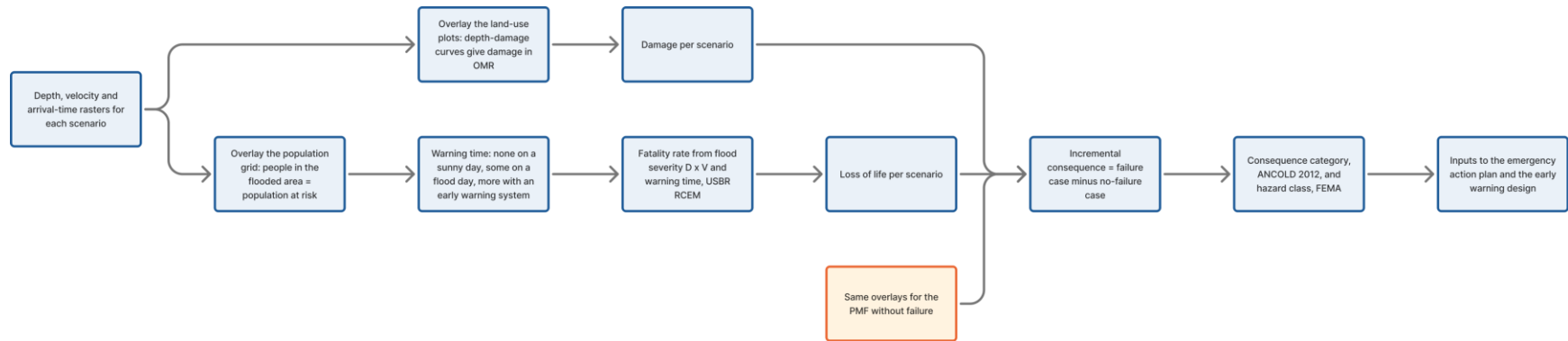


Figure 21 - From the model results to people at risk, loss of life, damage and the dam category.

9.1. POPULATION AT RISK

The population at risk of a run is the number of people inside its inundation extent. It is counted by overlaying the 2025 GHS-POP grid, which gives residents per 100 m cell, on the maximum depth raster, and summing the cells with a depth above 0.3 m. The count is made for the whole extent and for each hazard class, and separately for the incremental area of the flood-day run. The grid gives night-time residents; a daytime count, with schools, markets and the coast road traffic, will be made where the land-use layer identifies such uses, and both are reported.

9.2. LOSS OF LIFE

Loss of life is estimated with the Reclamation Consequence Estimating Methodology (USBR, 2015), the current empirical method used by the United States federal agencies. It assigns a fatality rate to each part of the population at risk from two things: how severe the flood is where those people are, measured by depth times velocity, and how much warning they had. The rate is read from the RCEM curves, which were fitted to sixty historical dam failures and floods, and multiplied by the people exposed:

$$LOL = \sum PAR_i \times f_i \quad (7)$$

where:

LOL estimated loss of life

PAR_i population at risk in zone i

f_i fatality rate for the flood severity and warning time of zone i (RCEM curves, 2015)

The warning time is the time between the moment people are told to leave and the arrival of the water. It depends on when the failure is noticed, how long the authorities take to issue the warning, and the arrival time from the maps. Table 23 gives the cases analysed. The sunny-day case is run with no warning, because a sudden failure gives none unless a warning system exists, and with the warning an early warning system would give; the difference is the benefit of that system, which the Client needs to decide its specification.



Table 23 - Warning-time cases for the loss-of-life estimate.

Case	Scenario	Warning assumed	Basis
W0	Sunny day	None: people learn of the flood when they see it	Sudden failure, no monitoring
W1	Sunny day	15 minutes at the town entry	Dam instrumentation alarm relayed by an early warning system
W2	Flood day	One hour	The PMF is a known cyclone event; the dam is watched and the town is already on alert
W3	Flood day	Three hours	Early warning system and an evacuation order issued during the storm

9.3. ECONOMIC DAMAGE

Direct damage to buildings and plots is computed with the depth-damage curves and the land-use classes of the flood-risk study of the scheme, applied to the maximum depth of each run. This keeps the dam break damage on the same basis as the flood damage already reported, so that the two can be compared. Damage to the dam itself, to the dikes and to roads and utilities is listed but not valued, because the values belong to the emergency planning rather than to the classification.

9.4. CRITICAL INFRASTRUCTURE AND SERVICES

The inundation extent of each run is intersected with the roads, the wadi crossings, the coast road, the power lines, the water and sewer networks, the schools, the health facilities and the mosques identified in the land-use and infrastructure layers, and the result is a list with the depth, the hazard class and the arrival time at each. The list is the input the emergency action plan needs to name shelters and routes.

9.5. INCREMENTAL CONSEQUENCES

For the flood-day scenario the consequences are reported twice: in total and as the increment over the no-failure run.

$$\Delta C = C_{failure} - C_{no\ failure} \quad (8)$$

where:



ΔC incremental consequence, for people at risk, loss of life or damage

C_{failure} consequence of the flood-day run

$C_{\text{no failure}}$ consequence of the PMF passing the dam intact

The sunny-day consequences are incremental by definition, because without the failure there is no flood.

9.6. CONSEQUENCE CATEGORY AND HAZARD CLASS OF THE DAM

The dam is classified in two systems, because the Client's counterparts use both. The ANCOLD guideline (2012) combines the population at risk with the severity of damage and loss into seven categories from Low to Extreme (Table 24). The FEMA hazard potential classification has three classes, and a dam whose failure would probably cause loss of life is High. The classification is made on the sunny-day and the incremental flood-day results, and the higher of the two governs.

Table 24 - Consequence categories for dams (ANCOLD, 2012). Severity of damage and loss is assessed on the infrastructure, business, community and environmental effects.

Population at risk	Severity: Minor	Severity: Medium	Severity: Major	Severity: Catastrophic
Less than 1	Very Low	Low	Significant	High C
1 to 10	Significant (Low if less than 5)	Significant	High C	High B
10 to 100	High C	High C	High B	High A
100 to 1,000	High B	High B	High A	Extreme
More than 1,000	High A	High A	Extreme	Extreme

9.7. INPUTS TO THE EMERGENCY PLAN AND THE WARNING SYSTEM

The analysis report will hand to the emergency action plan: the arrival-time maps and the times at the named places; the evacuation routes that stay dry or in class H1 for the longest; the shelters outside the H4 and higher areas; and the difference in loss of life between the warning cases, which is the case for the early warning system and sets the reaction time it must achieve.



10. SENSITIVITY, UNCERTAINTY AND QUALITY CHECKS

10.1. WHAT IS VARIED AND WHY

The breach parameters are the least certain input of the study, followed by the roughness and the sea level. Rather than take a single conservative value for each, which piles conservatism on conservatism and hides how much each one matters, the base case is run with the best estimates and each parameter is then varied on its own (Table 25). Each sensitivity run changes one thing from the base run it is compared with.

Table 25 - Sensitivity runs.

Parameter	Base	Low	High	Runs	What it tests
Breach width	86 m	43 m	129 and 156 m	S1-D, S2-D	Peak outflow and extent
Formation time	0.1 h	0.3 h	0.05 h	S1-D, S2-D	Peak and arrival time
Breach weir coefficient	1.66	1.44	-	S1-D	Peak
Roughness	Table values	Low end of range	High end of range	S1-D, S2-D	Depth, velocity, arrival time in the town
Mesh size	Chapter 7	-	Halved in the town and at the dam	S1-D	Numerical convergence
Sea level	Highest astronomical tide	Mean sea level	-	S2-D	Levels in the coastal part of the town
Starting pool, sunny day	84.5 m	Bottom outlet intake plus 10 m	-	S1-D	Reservoir level at the time of a failure
Failure timing, flood day	At the pool of 96.7 m	-	2 h after the peak	S2-D	Coincidence of breach and flood
Inflow flood, flood day	PMF	10,000-year	-	S2-D	Choice of the design flood
Reservoir representation	2D area	Storage area with the volume curve	-	S1-D	Drawdown effect on the peak
Parapet wall	Not credited	Credited to 96.7 m	-	S3-D	Overflow of the crest during the PMF
Dikes	In place / removed	-	-	All	Already part of the base runs

10.2. HOW THE UNCERTAINTY IS REPORTED

Each result is reported as its base value with the range from the sensitivity runs: the peak outflow, the arrival time at the town entry, the maximum depth at named places, the population at risk and the loss of life. The maps of the base case are the deliverable maps; the envelope of the maximum



depth over all runs is given as one additional map, so that the emergency plan can see the worst extent without the base maps being made pessimistic.

10.3. QUALITY CHECKS

Every run passes the checks of Table 26 before its results are used. A run that fails a check is corrected and rerun; the check log is delivered with the analysis report.

Table 26 - Quality checks applied to every model run.

Check	Criterion	When
Reservoir volume	2D area volume at FSL within 2 percent of the elevation-volume curve	Once, before the runs
Spillway rating	Model discharge at the PMF pool within 5 percent of the design rating	No-failure run
Routed pool	Peak pool within 0.2 m of the design team's routed level	No-failure run
Volume balance	Volume accounting error below 1 percent	Every run
Stability	No oscillation of the breach hydrograph; no cells with iteration warnings at the peak; smooth arrival-time map	Every run
Time step	Courant number at the wave front at or below 1; result unchanged when the maximum Courant is halved	Base runs
Mesh	Peak depth at the town entry changes by less than 5 percent when the mesh is halved	S1-D
Peak outflow	Between the Ritter and the weir hand estimates of Chapter 6, or the difference explained	Every failure run
Continuity to the sea	All water reaches the sea boundary or is stored on land; no water lost at the open edges	Every run
Consequence counts	Population and plot counts reproduced by an independent overlay in QGIS	Every base run
Documentation	Plan, geometry and unsteady file names, parameters and dates recorded in the run register	Every run

10.4. REVIEW

The model set-up and the base runs are reviewed by a senior engineer who did not build the model, against this methodology, before the sensitivity runs are made. The analysis report records the review and the changes it caused.



11. DELIVERABLES, WORK STEPS AND DATA NEEDED

11.1. DELIVERABLES

Table 27 - Deliverables of the dam break analysis.

Deliverable	Content	Format
Dam break analysis report	Method as approved here, model, results of all runs, consequences, classification, recommendations	Word and PDF, in this report format
Map atlas	General maps and map series of depth, hazard and arrival time for the six base runs, incremental map, envelope map	PDF, A3, and the QGIS project with all layouts
Rasters and vectors	Maximum depth, velocity, $D \times V$, arrival time, duration, hazard class for every run; inundation extents; reference-line hydrographs	GeoTIFF and shapefile, EPSG:32640
HEC-RAS model	Project with terrain, geometry, plans and results of every run, and the run register	HEC-RAS 6.6 project folder
Consequence tables	Population at risk, loss of life, damage and infrastructure lists per run and per warning case	Excel
Emergency planning inputs	Arrival times at named places, evacuation routes, shelter locations, warning-time benefit	Chapter of the report and GIS layers
Animations	Depth against time for the sunny-day and flood-day base runs	MP4

11.2. WORK STEPS AND PROGRAMME

Figure 22 shows the ten steps of the analysis and Figure 23 an indicative programme of about twelve weeks from the approval of this methodology and receipt of the data, of which the first two weeks depend on the design team.



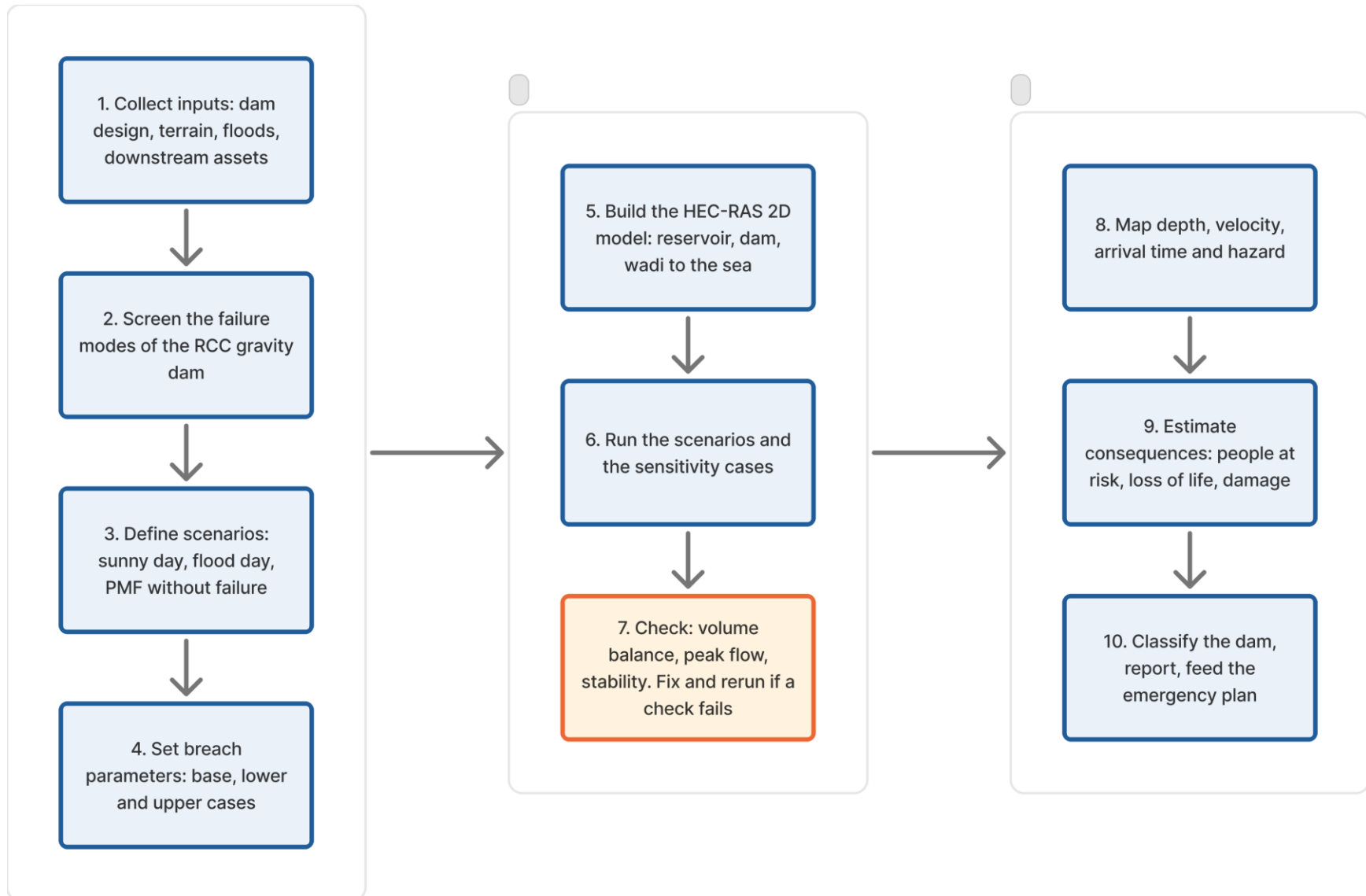


Figure 22 - The ten steps of the dam break analysis.



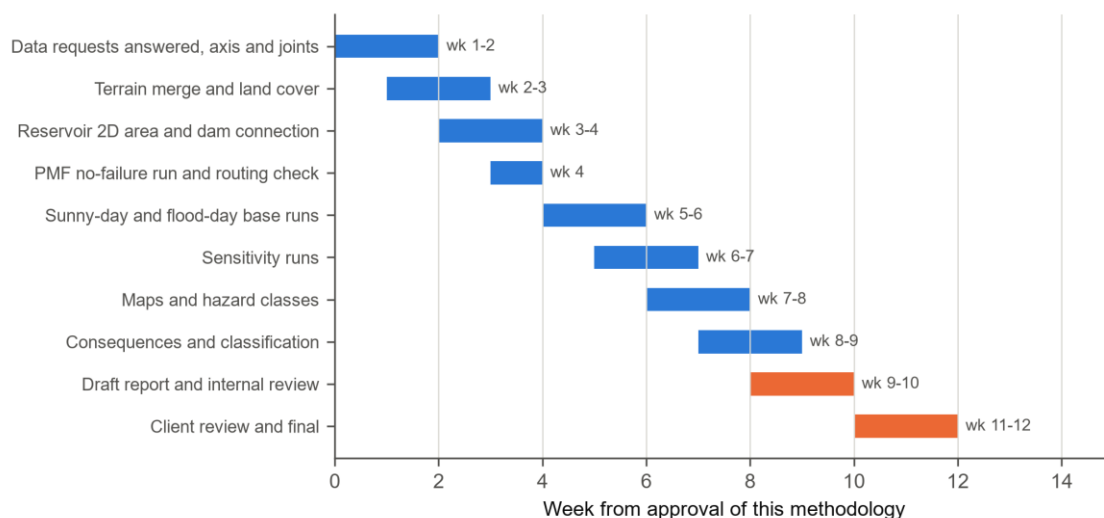


Figure 23 - Indicative programme of the analysis, in weeks from the approval of this methodology.

11.3. DATA STILL NEEDED

Table 28 lists the data that this methodology depends on and that are not yet in hand. Appendix D repeats the list as a register to be returned by the design team.

Table 28 - Data needed from the project team before the model is built.

Item	Needed for	From	Priority
Monolith joint layout with block widths and stations	Breach width in whole blocks	Dam design	Highest
Routed maximum water level during the PMF (96.0 or 96.7 m) and the routing calculation	Flood-day pool, routing check	Dam design / hydraulics	Highest
Elevation-area-volume table for the straight axis	Reservoir volume check	Dam design	High
Design surfaces of dam, spillway, stilling basin, bottom outlet chute, dikes, wadi crossings and roads	Terrain	Dam design / roads	High
Dam-site survey (5 cm) and drone DTM (2 m) files	Terrain	Survey	High
Spillway rating curve and design head of the ogee	Dam connection	Hydraulics	High
Tide levels at Qurayat (highest astronomical tide, mean sea level)	Sea boundary	National tide tables	Medium
Infrastructure layers: roads, utilities, schools, health facilities, mosques	Consequences	Municipality / project GIS	Medium
Daytime population indicators (schools, markets, workplaces)	Daytime population at risk	Municipality	Medium
Foundation and abutment failure mode notes from the dam safety review	Confirmation of the failure location	Geotechnics	Medium



11.4. DECISIONS REQUESTED FROM THE CLIENT

- Approval of the three scenarios and of the with-and-without-dikes pairing.
- Approval of the breach parameters of Chapter 6, in particular the base width of two spillway bays (86 m) and the formation time of 0.1 h.
- Confirmation that the parapet wall is not to be credited in the no-failure run.
- Confirmation of the warning-time cases of Chapter 9, which decide the benefit shown for the early warning system.
- Confirmation that the ANCOLD (2012) and FEMA classifications are the ones to report.



12. CONCLUSIONS AND RECOMMENDATIONS

The Wadi Majlas dam is a concrete gravity dam, and its dam break analysis must follow the rules for concrete gravity dams: a breach of whole monoliths with vertical sides, formed in minutes, with the number of failed blocks varied to cover the uncertainty. The regression equations written for embankment dams, used in earlier studies for the Client, do not apply and would give a slower, smaller flood and a longer warning time than the dam can justify.

Three scenarios cover what the dam can do: a sunny-day failure at full supply level, a flood-day failure at the peak pool of the probable maximum flood, and the probable maximum flood passing the dam intact. Each is run with and without the downstream dikes. The flood is computed in HEC-RAS 2D with the full shallow water equations on a model that runs from the reservoir to the sea, built on the existing model of the downstream reach. The results are maps of depth, velocity, arrival time and hazard, the population at risk, an estimate of loss of life for stated warning times, the damage, and the consequence category of the dam.

The Consultant recommends that the Client approves this methodology as the basis of the analysis, and that the design team supplies the four items on which the model depends: the monolith layout, the routed maximum water level, the reservoir curve for the straight axis, and the design surfaces. With these in hand the analysis can be completed in about twelve weeks.

Two points deserve the Client's attention now rather than at the end. The parapet wall at 96.7 m is the only thing between the PMF pool and the crest at 95.5 m; this methodology does not credit it, and the design team may wish to confirm the freeboard basis. And the sunny-day case with no warning will give the largest loss of life of the study; the difference between it and the case with fifteen minutes of warning is the strongest argument for the early warning system, and the Client may wish to decide the scope of that system on this evidence.



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APPENDIX A - EMBANKMENT BREACH EQUATIONS, FOR REFERENCE ONLY

The equations below are the standard tools for embankment dams. They are reported here so that the reader can see what they would give for Majlas and why they are not used for a concrete gravity dam (Chapter 6). The values are computed for the sunny-day pool: a volume of 102.2 Mm³ and a height of water above the breach floor of 61.5 m.

Froehlich (2008)

$$B_{avg} = 0.27k_o V_w^{0.32} h_b^{0.04} \quad (9)$$

where:

B_{avg} average breach width (m)

k_o 1.3 for overtopping, 1.0 for piping

V_w volume of water above the breach floor at failure (m³)

h_b height of the breach (m)

$$t_f = 63.2 \sqrt{\frac{V_w}{gh_b^2}} \quad (10)$$

where:

t_f formation time (s)

g 9.81 m/s²

For Majlas these give an average width of 116 m (piping) to 151 m (overtopping) and a formation time of 0.92 hours, with side slopes of 0.7 (piping) to 1.0 (overtopping) horizontal to vertical. The width is of the same order as the upper case of Chapter 6, but the time is about 9 times longer than the concrete dam value, which is why the equation would understate the peak.



MacDonald and Langridge-Monopolis (1984)

$$V_{er} = 0.0261(V_{out}h_w)^{0.769} \quad (11)$$

where:

V_{er} volume of embankment eroded (m^3), earthfill dams

V_{out} volume of water released (m^3)

h_w depth of water above the breach floor (m)

$$t_f = 0.0179V_{er}^{0.364} \quad (12)$$

where:

t_f formation time (h)

For Majlas: an eroded volume of about 0.89 Mm^3 and a formation time of 2.6 hours. The breach width follows from the eroded volume and the dam section, and has no meaning for a concrete dam.

Von Thun and Gillette (1990)

$$B = 2.5h_w + C_b \quad (13)$$

where:

B average breach width (m)

h_w depth of water above the breach floor (m)

C_b 54.9 m for reservoirs larger than 12.3 Mm^3

With a formation time of $0.015 h_w$ for easily eroded material and $0.020 h_w + 0.25$ for erosion-resistant material. For Majlas: a width of 209 m and a time of 0.92 to 1.48 hours.



Xu and Zhang (2009)

Xu and Zhang fitted multiplicative equations to 182 earth and rockfill failures in which the breach width, depth and time depend on the dam height, the reservoir shape, the dam type (corewall, concrete face, homogeneous), the failure mode and the erodibility of the fill. The erodibility term dominates the result and cannot be defined for a concrete dam. The equations are not evaluated here.

Table 29 - Embankment breach equations evaluated for the Majlas sunny-day pool, against the values adopted.

Equation	Fitted to	Width for Majlas (m)	Time for Majlas (h)	Applicable to a gravity dam?
Froehlich (2008)	74 embankment failures	116 to 151	0.92	No
MacDonald and Langridge-Monopolis (1984)	42 embankment failures	From eroded volume	2.6	No
Von Thun and Gillette (1990)	57 embankment failures	209	0.92 to 1.48	No
Xu and Zhang (2009)	182 earth and rockfill failures	Not evaluated	Not evaluated	No
Federal guidance for concrete gravity dams (Chapter 6)	Concrete dam case histories	43 to 156	0.05 to 0.3	Yes



APPENDIX B - HEC-RAS INPUT SHEET

This sheet lists every input the modeller must set in HEC-RAS 6.6 for the base runs, where it is entered, and where its value comes from. It is the checklist for building the model and for the review of Chapter 10.

Table 30 - HEC-RAS input sheet for the base runs.

Where in HEC-RAS	Input	Value or source
RAS Mapper > Terrains	Merged terrain	Chapter 7, Table of terrain sources; priority order as listed
RAS Mapper > Map Layers > Land Cover	Land cover layer and Manning's n table	Chapter 7, Manning's n table
Geometry > 2D Flow Areas	Reservoir area; downstream area; cell sizes; refinement regions; breaklines	Chapter 7, mesh table
Geometry > 2D Flow Areas > Land cover override	Roughness override where the layer is wrong	Only with a note in the run register
Geometry > SA/2D Area Connections	Connection along the dam axis; weir profile; weir shape and coefficients; weir width	Chapter 7, dam connection table
Geometry > Reference Lines	Gorge exit, town entry, coast road, shoreline	Chapter 8
Unsteady Flow > Boundary Conditions	PMF flow hydrograph at the reservoir; tributary hydrographs; sea stage; normal depth edges	Chapter 7, boundary condition table; hydrographs from HEC-HMS
Unsteady Flow > Initial Conditions	Reservoir 2D area initial water surface 84.5 m; downstream dry; sea at tide level	Chapter 7
Plan > Breach (plan data)	Breach method: user entered; centre station; final bottom width; final bottom elevation; side slopes; weir coefficient; formation time; failure mode; trigger; progression	Chapter 6 table of breach parameters
Plan > Computation Options and Tolerances > 2D	Equation set SWE-EM; theta 1.0; tolerances 0.003 m; iterations 20; turbulence default	Chapter 7, computation settings
Plan > Unsteady Computation Options > Advanced Time Step	Adaptive on Courant; maximum 1.0; halve above, double below 0.5; initial 1 s	Chapter 7
Plan > Simulation time window	Sunny day: 2 h warm-up + 12 h; flood day: 60 h	Chapter 7
Plan > Output options	Hydrograph interval 10 s; mapping interval 1 min then 5 min; detailed output	Chapter 7
Plan > Restart file	Write at the end of the warm-up; read for sensitivity runs	Chapter 7
RAS Mapper > Results	Max depth, velocity, D×V, arrival time (0.3 m), duration, hazard (AIDR); export GeoTIFF 2 m	Chapter 8



APPENDIX C - MAP LAYOUT AND SYMBOLOGY

All maps of the analysis use the print layout of the project GIS: A4 landscape at 300 dpi, title and subtitle at the top left, legend at the left, north arrow and scale bar at the top right, and an information table at the bottom right that carries the run identifier, the scenario, the breach parameters and the model date. The four maps of this report were produced from that layout and are saved in the QGIS project of the working folder for reuse. Table 31 fixes the symbology so that every map of the study reads the same way.

Table 31 - Symbology of the map series.

Layer	Symbology
Maximum depth	Sequential blues in classes 0.3, 0.5, 1.0, 2.0, 4.0, 8.0 m and above; below 0.3 m not shown
Hazard class	H1 to H6 in the six-step blue ramp of the flood-risk study, from light to dark
Arrival time	Classes 0-15, 15-30, 30-60, 60-120, 120-240 minutes and above, from red to yellow to green
Velocity	Sequential oranges in classes 0.5, 1.0, 2.0, 4.0 m/s and above
Incremental depth	Diverging: blue where the failure adds depth above 0.6 m, grey below
Dam axis	Red line, 1.6 mm
Dikes	Orange line, 1.1 mm
Land use	Residential pink, commercial yellow, industrial violet, agricultural green
Population grid	Blue classes 1-5, 5-20, 20-50, 50-100, over 100 persons per cell
Base map	Satellite image; hillshade of the terrain for the reach maps



APPENDIX D - DATA REQUEST REGISTER

The register below is to be returned by the design team with the file names and dates of the items supplied.

Table 32 - Data request register.

No.	Item	Format requested	Needed by	Supplied (file, date)
1	Monolith joint layout: stations and widths of blocks B1 to B17	Drawing or table	Week 1	
2	Routed maximum water level during the PMF and the routing calculation (96.0 or 96.7 m)	Table and note	Week 1	
3	Elevation-area-volume table for the straight axis	Excel	Week 1	
4	Design surfaces: dam, spillway, stilling basin, bottom outlet chute, dikes, wadi crossings, roads	DWG or DTM, EPSG:32640	Week 2	
5	Dam-site survey (5 cm) and drone DTM (2 m)	GeoTIFF, EPSG:32640	Week 2	
6	Spillway rating curve and ogee design head	Table	Week 2	
7	Tide levels at Qurayat: highest astronomical tide, mean sea level	Note with source	Week 3	
8	Infrastructure layers: roads, utilities, schools, health facilities, mosques	Shapefile	Week 4	
9	Daytime population indicators	Table or shapefile	Week 4	
10	Foundation and abutment failure mode notes	Note	Week 3	

